

Fractional Brownian Motion: Markovian
Approximation and Applications to Rough
Volatility Modelling

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Notation

$\mathbb{N}$	natural numbers $\{1, 2, 3, \dots\}$
$\mathbb{R}^+$	non-negative reals $[0, \infty)$
$X \stackrel{d}{=} Y$	equality in distribution of random variables
I_a^α	left fractional integral of order α over $[a, b]$
D_a^α	left fractional derivative of order α over $[a, b]$
B_t^H	Fractional Brownian motion with hurst parameter H
I_m	$m \times m$ identity matrix
$\mathbf{A}^T$	transpose of the matrix $\mathbf{A}$
$diag(a_1, \dots, a_m)$	diagonal matrix with entries $a_1, \dots, a_m$
$x_n \rightarrow a$	x_n The series converges to a as $n \rightarrow \infty$
$x_n \downarrow a$	x_n converges to a from above
$x_n \uparrow a$	x_n converges to a from below
$x \wedge y$	minimum of x, y
$x \vee y$	maximum of x, y
B_t^H	fractional Brownian motion with Hurst parameter H at time t
Z_t^β	standard Ornstein-Uhlenbeck process with mean reversion β at time t

Abstract

In this report we introduce fractional Brownian motion, adapt the lifted Heston model from [10] to construct an analogous lifted fBm model, prove convergence for the $H < \frac{1}{2}$ case, and give strong numerical validation for the $H > \frac{1}{2}$ case. We then situate these findings within the modern rough volatility literature, connecting our construction to the lifted Heston model it was adapted from.

0.1 Introduction

Fractional Brownian motion (fBm_H) has emerged as an indispensable tool in modern quantitative finance, particularly in the characterization of asset price volatility. Despite its widespread adoption by both theoretical mathematicians and financial practitioners, there remains a pronounced pedagogical gap: accessible, self-contained treatments of fBm_H tailored to an advanced undergraduate level are notably scarce. Existing literature (see [18], [11], [16], [14]) typically falls into one of two categories. They are either couched in a degree of abstraction far exceeding the needs of a practitioner, or they introduce key integral representations axiomatically, without providing the underlying constructive intuition. This project aims to bridge this gap by providing a rigorous yet accessible treatment of fBm_H , developing its integral representations naturally from core principles.

The mathematical centrepiece of this work is the theory underlying the infinite-dimensional Ornstein-Uhlenbeck (OU) representation for fractional Brownian motion. Building on this framework, and adapting the node and weight parametrisation underlying the lifted Heston model of [10], we construct a finite-dimensional Markovian approximation of fBm_H itself – *lifted fBm_H*, the central original contribution of this report. For the sub-diffusive regime $H < \frac{1}{2}$, we prove that this approximation converges to true fBm in $L^2(\Omega)$; for the super-diffusive regime $H > \frac{1}{2}$, where the same proof technique fails for reasons we identify precisely, we instead give strong numerical evidence that convergence continues to hold. Throughout, our approach is not merely to construct an approximation but to validate it quantitatively against the known fBm covariance, and to characterise precisely where and why a naive discretisation fails to converge – motivating the optimised quadrature schemes used in practical applications such as rough Heston [10].

To demonstrate the practical relevance of this theory, Chapter 3 illustrates how fBm_H has revolutionized contemporary volatility modelling. We introduce the classical Heston model alongside its rough counterpart, the rough Heston model. We then focus on an active area of contemporary research: the *lifted Heston model*. This framework utilizes the same Markovian lifting technique developed in Chapter 1 to construct a tractable, finite-dimensional Markovian approximation to the otherwise non-Markovian rough Heston model.

The remainder of this report is structured as follows. Chapter 1 develops the foundational theory of fBm_H from first principles, establishing its core properties before naturally deriving its integral representations – beginning with the classical Mandelbrot-Van Ness representation and culminating in the infinite-dimensional OU representation – before constructing, validating, and proving convergence of lifted fBm_H . Chapter 3 applies this theoretical machinery to financial volatility modelling, drawing on contemporary literature to provide an exposition of the lifted Heston model. We explicitly note that a rigorous treatment of stochastic differential equations directly driven by fBm_H falls outside the scope of this text; instead, our focus remains centred on the mechanics and validation of the Markovian approximation.

Chapter 1

Fractional Brownian Motion

1.1 Preliminaries

1.1.1 Fractional Calculus

Before going into the definition & representations of fBm, it will be heuristically important for us to first understand fractional calculus. The idea behind fractional calculus is to give a sensible definition for what it means to have an integral or derivative of fractional order. Following [13] we begin by looking at the Cauchy formula for iterated integration.

Theorem 1.1.1 (Cauchy n -Fold Integral Formula). *Given any $a, x \in \mathbb{R}$ and an integrable function f , the following formula holds for all $n \in \mathbb{N}$*

$$(I_a^n f)(x) := \int_a^x \int_a^{x_1} \cdots \int_a^{x_{n-1}} f(x_{n-1}) dx_{n-1} \cdots dx_1 dx = \frac{1}{(n-1)!} \int_a^x (x-t)^{n-1} f(t) dt$$

Proof. We prove this by induction on n . The base case $n = 1$ is trivial. Suppose that for $n - 1$ we have

$$(I_a^{n-1} f)(x) = \frac{1}{(n-2)!} \int_a^x (x-t)^{n-2} f(t) dt.$$

Then by definition we have

$$\begin{aligned}
(I_a^n f)(x) &= \int_a^x (I_a^{n-1} f)(s) ds \\
&= \frac{1}{(n-2)!} \int_a^x \int_a^s (s-t)^{n-2} f(t) dt ds \\
&= \frac{1}{(n-2)!} \int_a^x f(t) \int_t^x (s-t)^{n-2} ds dt \quad (\text{By Fubini's theorem}) \\
&= \frac{1}{(n-2)!} \int_a^x f(t) \cdot \frac{(x-t)^{n-1}}{n-1} dt \\
&= \frac{1}{(n-1)!} \int_a^x (x-t)^{n-1} f(t) dt.
\end{aligned}$$

□

Now keeping in mind that the gamma function ¹ satisfies $\Gamma(n) = (n-1)!$ for all non-negative integers, it seems that using this would give natural extension of our integral formula to the positive reals [13].

Definition 1.1.1 (Left Hand Riemann-Liouville Fractional Integral). Let $I = [a, b]$ be a finite interval on the real axis $\mathbb{R}$. The left-side fractional integral $I^\alpha f$ of order² $\alpha \in \mathbb{R}^+$ is defined for $x > a$ by

$$(I_a^\alpha f)(x) = \frac{1}{\Gamma(\alpha)} \int_a^x \frac{f(t)}{(x-t)^{1-\alpha}} dt$$

Now by noting that we want derivatives and integrals to be “inverses” of one another, we expect any reasonable definition of the fractional derivative to satisfy $(D^\alpha(I^\beta f))(x) = (I^{b-\alpha} f)(x)$.

Definition 1.1.2 (Left Hand Riemann-Liouville Fractional Derivative). The left-side fractional derivative $D^\alpha f$ of order $\alpha \in \mathbb{R}$ on $[a, b]$ is defined by

$$\begin{aligned}
(D_a^\alpha f)(x) &= \left(\frac{d}{dx} \right)^n (I_a^{n-\alpha} f)(x) \\
&= \frac{1}{\Gamma(n-\alpha)} \left(\frac{d}{dx} \right)^n \int_a^x \frac{f(t) dt}{(x-t)^{\alpha-n+1}}.
\end{aligned}$$

Where we take any natural $n > \alpha$, thus the standard is to use $n = [\alpha] + 1$ which is the smallest integer larger than α .

¹ $\Gamma(x) := \int_0^\infty t^{x-1} e^{-t} dt$

²Integrals of complex order can also be defined but we choose to omit that here as it is not relevant to our work, although is interesting in its own right.

1.1.2 Hölder Continuity

If we want to look at continuous time process it will be important that we have a way of measuring how “rough” the sample paths are. This motivates the idea of Hölder continuity.

Definition 1.1.3 (Hölder Continuity [7]). Let $0 < \gamma \leq 1$. A function $f : [0, T] \rightarrow \mathbb{R}$ is called uniformly Hölder continuous with exponent $\gamma > 0$ if there exists a constant K such that

$$|f(t) - f(s)| \leq K|t - s|^\gamma \quad \text{for all } s, t \in [0, T].$$

We say f is Hölder continuous with exponent $\gamma > 0$ at the point s if there exists a constant K such that

$$|f(t) - f(s)| \leq K|t - s|^\gamma \quad \text{for all } t \in [0, T].$$

From a heuristic point of view, γ tells us how much the functions oscillate over a small interval. Using sufficiently small time intervals $|t - s| = \delta > 0$ it follows that δ^γ increases as γ decreases. With $\gamma = 1$ giving us a Lipschitz function, which is known to be differentiable almost everywhere³, as $\gamma \rightarrow 0$ the bound on oscillation becomes independent of $|t - s|$, meaning the function need not be continuous at all.

We will see that the sample paths of fBm are Hölder continuous with exponent $\gamma < H$, where H is the Hurst parameter, making this a natural tool for measuring path roughness. A good general theorem which we will later use to prove Hölder continuity of fBm paths is this important theorem of Kolmogorov of which we do not include a proof of here.

Theorem 1.1.2 (Kolmogorov-Chentsov continuity theorem [18]). *Assume that for a stochastic process $(X_t)_{t \geq 0}$ there exist $K > 0, p > 0, \beta > 0$ such that for all $t \geq 0, s \geq 0$*

$$\mathbb{E}[|X_t - X_s|^p] \leq K|t - s|^{1+\beta}.$$

Then the process X has a continuous modification⁴. Moreover, for any $\gamma \in (0, \beta/p)$ and $T > 0$ the modified process $\tilde{X}$ is γ -Hölder continuous on $[0, T]$.

Heuristically this theorem just states the implications between the Hölder regularity of the p -th moment and the random itself.

We lastly show that fractional integration increases Hölder continuity by order of integral to motivate time representation of fBm in 1.3.

Theorem 1.1.3 (Integration increases Hölder regularity). *Suppose $0 < \gamma \leq 1$ and $\alpha \in \mathbb{R}$ satisfy $0 < \gamma + \alpha \leq 1$. Further suppose that $f : [a, b] \rightarrow \mathbb{R}$ is γ -Hölder continuous, then $I^\alpha f$ is $(\gamma + \alpha)$ -Hölder continuous.*

This proof uses standard estimates which we choose to omit (see [17]).

³This follows from Rademacher's theorem, which states that any Lipschitz function is differentiable almost everywhere (see [7]).

⁴i.e. a process $\{\tilde{X}_t, t \geq 0\}$ such that $\tilde{X} \in C[0, \infty)$ and for all $t \geq 0$ $\mathbb{P}(X_t = \tilde{X}_t) = 1$.

1.2 Fractional Brownian Motion

The fruits of fractional calculus will now pay off as it is an essential tool used in representing fBm as we will now come to see.

1.2.1 Definition & Motivation

The assumptions of independent, stationary increments along with continuous paths give Brownian motion. Now suppose we remove the dependent increments property and want to keep self-similarity. For this we will now define a new process - fBm - for which we will retain the assumptions of stationary increments and continuous paths, but will allow the increments to be dependent. The process B_t^H we construct depends on a parameter $H \in (0, 1)$ called the Hurst index. It measures the correlation of the increments. We will also assume that the process is self-similar⁵

- If $a > 0$, then the distribution of

$$X_t := a^{-H} B_{at}^H$$

is the same as that of B_t^H . In particular, B_t^H has the same distribution as $t^H B_1^H$.

Our final assumption is that the process is a centred Gaussian process.

- For every $t_1, \dots, t_k$, the random vector $(B_{t_1}^H, \dots, B_{t_k}^H)$ has a joint normal distribution with mean zero.

We normalize so that $\text{Var}(B_1^H) = \mathbb{E}[(B_1^H)^2] = 1$ and hence

$$\text{Var}(B_t^H) = \text{Var}(t^H B_1^H) = t^{2H} \text{Var}(B_1^H) = t^{2H}.$$

Thus since we would want increments to be stationary⁶ it seems reasonable to have

$$\begin{aligned} \mathbb{E}[(B_t^H - B_s^H)^2] &= |t - s|^{2H} \\ \mathbb{E}[(B_t^H)^2] + \mathbb{E}[(B_s^H)^2] - 2\mathbb{E}[B_t^H B_s^H] &= |t - s|^{2H}. \end{aligned}$$

This motivates the covariance structure upon which we will now define fBm.

Definition 1.2.1 (Fractional Brownian Motion). If $H \in (0, 1)$, the fractional Brownian motion with Hurst parameter H (fBm _{H}) is the centred (mean zero) Gaussian process B_t^H with continuous paths such that for all $s, t \geq 0$

$$\text{Cov}(B_s^H, B_t^H) = \mathbb{E}[B_s^H B_t^H] = \frac{1}{2} [s^{2H} + t^{2H} - |s - t|^{2H}].$$

If $H = 1/2$, then fractional Brownian motion is the same as usual Brownian motion.

⁵Any self-similar process must be of this form, since if $X_{at} = \lambda X_t$ then it follows that $X_{at} X_{a't} = \lambda \lambda' X_t$ which clearly gives us a multiplicative functional equation which can easily be shown to only have solutions satisfying a power law.

⁶A stochastic process is stationary if it's distribution is the same at all times, thus for a centred Gaussian process it suffices that the variance is constant.

We will prove in then next section 1.3 using an integral representation that such a process actually exists, but for now we assume it does exist. We next state some theorems to build up some intuition on the behaviour of fBm.

Theorem 1.2.1 (Covariance Between Increments). *If $0 \leq s < t < \infty$, note that*

$$\begin{aligned}\mathbb{E} [B_s^H (B_t^H - B_s^H)] &= \mathbb{E} [B_s^H B_t^H] - \mathbb{E} [(B_s^H)^2] \\ &= \frac{1}{2} [t^{2H} - s^{2H} - (t-s)^{2H}]\end{aligned}$$

This covariance structure intuitively can be explained by the following idea on how rough the paths are for each H . Namely since $B_{t+\delta}^H - B_t^H \sim N(0, \delta^{2H})$, we can write roughly

$$|B_{t+\delta}^H - B_t^H| \approx \delta^H$$

In other words, the Hölder exponent of fBm_H is given by the Hurst index (just under) H .

If $H > 1/2$, the paths are “smoother” than Brownian paths and if $H < 1/2$, the paths are “rougher”. More formally we have the following theorem.

Theorem 1.2.2 (Hölder Continuity of fBm_H). *If $(B_t^H)_{t \geq 0}$ is a fBm_H process then it is γ -Hölder continuous for all $\gamma \in (0, H)$.*

Proof. We use the Kolmogorov–Chentsov continuity theorem 1.1.2 as follows. Fix $t > s$. Since B^H is Gaussian and centered, we have that

$$B_t^H - B_s^H \stackrel{d}{=} \sqrt{\mathbb{E} [|B_t^H - B_s^H|^2]} Z,$$

where $Z \sim N(0, 1)$. So by recalling that $\mathbb{E} [|B_t^H - B_s^H|^2] = |t - s|^{2H}$ then we have

$$B_t^H - B_s^H = |t - s|^H Z$$

So given any $p \in \mathbb{N}$ we have that

$$\mathbb{E} [|B_t - B_s|^p] = \mathbb{E} [|Z|^p] |t - s|^{pH}$$

We deduce from the Kolmogorov–Chentsov lemma (Theorem 1.1.2) by noting that $K_p = \mathbb{E} [|Z|^p]$ is constant for each p and by taking $\beta = pH - 1$ the result follows for all $0 < \gamma < \frac{\beta}{p} = H - \frac{1}{p}$. Hence by taking p so that $pH > 1$ we have that B_t^H has Hölder regularity γ for all $\gamma \in (0, H)$. $\square$

1.2.2 Correlation & Long-Range Dependency

A fBm_H is clearly not a stationary process (variance of increments changes over time). However it was baked into the way we defined fBm_H in Section 1.2.1 that the increment process is stationary which allows us to talk about the autocorrelation between increments of arbitrary times. In this subsection we will

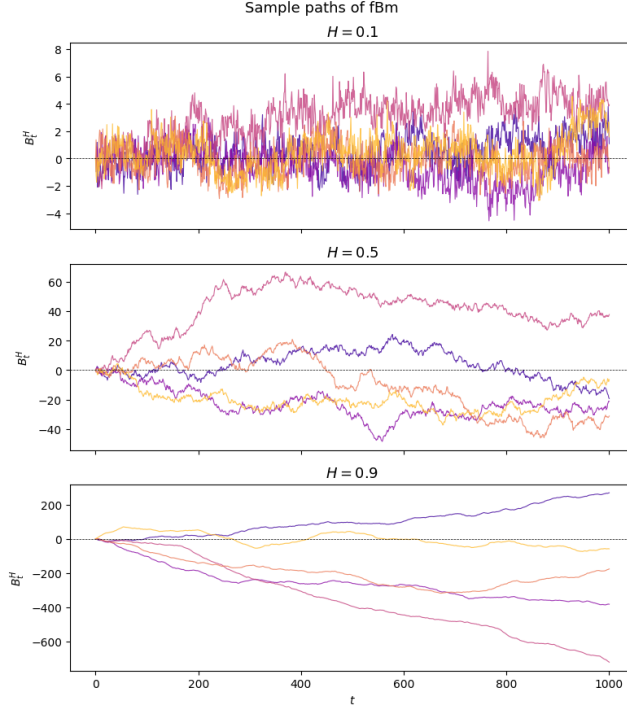


Figure 1.1: Sample paths of fractional Brownian motion for $H = 0.1, 0.5, 0.9$. $H = 0.5$ recovers standard Brownian motion (independent increments); $H = 0.1$ gives negatively correlated increments and a rougher path; $H = 0.9$ gives positively correlated increments and a smoother, more persistent path. Implementation available at [5]

show some properties of the dependence of the increments of fBm, and also exhibit when the fBm has long-range dependence (long-memory).

We will start by looking at the autocorrelation between increments. Consider the following discrete-time Gaussian process $X_n = B_n^H - B_{n-1}^H, n \in \mathbb{N}$, known as a fractional Gaussian noise (fGn), a generalisation of a traditional white noise corresponding to the case where $H = \frac{1}{2}$. Here X_k is a sequence of $N(0, 1)$ random variables that are not necessarily independent of one another.

If we consider $\rho_n = \mathbb{E}(X_{k+n}X_k)$, we can observe that it depends only on n , and not k . We have that

$$\begin{aligned} \rho_n &= E(X_{k+n}X_k) = E((B_{k+n}^H - B_{k+n-1}^H)(B_k^H - B_{k-1}^H)) \\ &= \frac{1}{2} [(n+1)^{2H} - n^{2H} - (n^{2H} - (n-1)^{2H})]. \end{aligned}$$

We now pose the question as to what this autocorrelation structure means for fBm_H over long periods of time. Heuristically long range dependence (LRD) is

just the idea that a stochastic process' current value can have long term effects on the future value, or in other words the increments between two distant values may have sufficiently large correlation. Many different definitions for LRD are used in the literature, we will use the following to define LRD:

Definition 1.2.2 (Long Range Dependence). Suppose $(X_t)_{t \geq 0}$ is a stationary process and let $\rho_n := Cov(X_t, X_{t+n})$ which is independent of t since the process is stationary. We say that X has the long range dependence property if

$$\sum_{n=1}^{\infty} \rho_n = \infty.$$

The motivation for this definition is rooted in statistical inference where this sum is important when computing statistics such as the sample mean or variance.

Theorem 1.2.3 (Long range dependence of fBm_H). *fBm_H has long range dependence for $H \in (\frac{1}{2}, 1)$ and short range dependence for $H \in (0, \frac{1}{2})$.*

Proof. We prove this by first noting the Taylor expansion of $(1+x)^{2H}$ for small $x = \frac{1}{n}$ allowing us to rewrite ρ_n as follows.

$$\begin{aligned} \rho_n &= \frac{1}{2} n^{2H} \left[\left(1 + \frac{1}{n}\right)^{2H} + \left(1 - \frac{1}{n}\right)^{2H} - 1 - 1 \right] \\ &= \frac{1}{2} n^{2H} \left[\left(1 + 2H \frac{1}{n} + \frac{1}{2} \cdot 2H(2H-1) \frac{1}{n^2}\right) \right. \\ &\quad \left. + \left(1 - 2H \frac{1}{n} + \frac{1}{2} \cdot 2H(2H-1) \frac{1}{n^2}\right) - 1 - 1 + O\left(\frac{1}{n^3}\right) \right] \\ &= H(2H-1) n^{2H-2} + O(n^{2H-3}) \end{aligned}$$

Now first notice that since $2H-3 < -1$ for all $H \in (0, 1)$ it does not effect the convergence of our sum⁷. And for $H \in (\frac{1}{2}, 1)$ it follows that $2H-2 > -1$ so

$$\sum_{n=0}^{\infty} \rho_n = \infty \quad \text{for } H > \frac{1}{2}$$

and similarly this series converges for $H < \frac{1}{2}$. We note also that the $H = \frac{1}{2}$ case is just our classical Brownian motion which has $\rho_n = 0$ so we have short range dependence too. $\square$

1.2.3 Semi-Martingale Property

In this section we very briefly demonstrate that fBm_H is not integrable in the sense of the classic stochastic Ito (or Stratonovich integral) integral. We leave out

⁷This follows from the p -test for convergent series which states that $\sum n^p$ converges if and only if $p < -1$ and noting that adding a convergent term to the inside of our series doesn't effect it's convergence

the details so as to keep the main idea in mind going forward to the mathematical core in the next section.

Definition 1.2.3 (Semimartingale). A process $X = X_t, t \geq 0$ with associated filtration $\mathcal{F}_t$ is called a semimartingale if it admits the following representation:

$$X_t = X_0 + M_t + Y_t,$$

where X_0 is $\mathcal{F}_0$ -measurable, M is a $\mathcal{F}_t$ -local martingale with $M_0 = 0$, and Y_t is a process of locally bounded variation.

Lemma 1.2.4. A semimartingale X_t has bounded quadratic variation.

With this at hand, we can address the non-semimartingale property of fBm.

Theorem 1.2.5. Let B_t^H be a fBm with Hurst index $H \in (0, 1)$. Then, for $H \neq \frac{1}{2}$, B_t^H is not a semimartingale.

The proof of this theorem follows by demonstrating that fBm_H has unbounded quadratic variation.

1.2.4 Markov Property

A stochastic process is said to be a Markov process if its values at time s encapsulates all relevant future information.

Definition 1.2.4 (Markov property). A stochastic process $(X_t)_{t \geq 0}$ is said to be a Markov process if

$$X_t | \mathcal{F}_s \stackrel{d}{=} X_t | X_s$$

where $\mathcal{F}_s = \sigma(X_u : u \leq s)$ is the sigma algebra generated by X up to time s , also known as the history of X up to time s .

One of the key problems that makes fBm_H difficult to work with, especially when it comes to financial modelling, is the fact that it is non-markovian. In section 1.4.4 we will demonstrate a slick workaround for this non-markovian nature, thus to motivate the necessity of the representation 1.4.4 we prove here this non-markovian property. The following proof is an original and highly simplified version of that given in [16].

Theorem 1.2.6 (fBm_H is non-Markovian [16]). For $H \neq \frac{1}{2}$ fBm_H is not a markov process

Proof. Suppose, for contradiction, that B_t^H is Markovian. The following identity for the Gaussian Markovian random variable holds (see [16]):

$$E [B_s^H B_u^H] E [(B_t^H)^2] = E [B_s^H B_t^H] E [B_t^H B_u^H].$$

for any $0 \leq s \leq t \leq u$. Hence for $s = 1, t = 2, u = 3$ we get

$$\begin{aligned} E[B_1^H B_3^H] E[(B_2^H)^2] &= E[B_1^H B_2^H] E[B_2^H B_3^H] \\ \implies \frac{1}{2}[1^{2H} + 3^{2H} - 2^{2H}] \times 2^{2H} &= \frac{1}{2}[1^{2H} + 2^{2H} - 1^{2H}] \times \frac{1}{2}[2^{2H} + 3^{2H} - 1^{2H}] \\ \implies 2^{2H} - 3^{2H-1} - 1 &= 0 \end{aligned}$$

Now we can show⁸ that $H = \frac{1}{2}$ is the unique solution to the above equation for $H \in (0, 1)$. Hence we have a contradiction of our Markovian assumption for $H \neq \frac{1}{2}$. The $H = \frac{1}{2}$ case is simply standard Brownian motion which is known to be Markovian. $\square$

1.3 Mandelbrot-Vann Ness representation

So we have seen that fBm_H is a Gaussian process that is (just under) H -Hölder continuous. Moreover we have seen in section 1.1.1 that fractional integration increases the Hölder parameter of our integrand by the order of integration. Thus it seems natural to try to directly represent fBm_H via an integral in this way. We start by noting from [7] that we can represent a standard Brownian motion $(B_t)_{t \geq 0}$ by the stochastic integral

$$B_t = \int_0^t dB_s.$$

This heuristically implies that our increment term dB_s has Hölder regularity $-\frac{1}{2}$. Hence if we fractionally integrate dB_s by order $H + \frac{1}{2}$ then we may result in a Gaussian process that is H -Hölder continuous which is exactly what we are looking for. Thus we may be tempted to formally define some process⁹

$$X_t = (I_0^{H+\frac{1}{2}} dB)(t) = \frac{1}{\Gamma(H + \frac{1}{2})} \int_0^t (t-s)^{H-\frac{1}{2}} dB_s.$$

However this does not have the stationary increments so cannot possibly be fBm_H. The process defined above was first introduced by Paul Lévy [14] and is known as Louville fBm or type 2 fBm. This process is explored more in [12] but this integral puts too great an importance on the origin for many applications.

In order to fix these issues it was suggested by Mandelbrot and Van-Ness in [14] that we should instead integrate from $-\infty$ to ensure that we have the stationary increments that we want, then subtract our Lévy representation to ensure that the process starts at the origin. Formally this suggests that fBm may be described by a stochastic process of the form:

$$\begin{aligned} X_t &= (I_{-\infty}^{H+\frac{1}{2}} dB)(t) - (I_0^{H+\frac{1}{2}} dB)(t) \\ &= \frac{1}{\Gamma(H + \frac{1}{2})} \left(\int_{-\infty}^0 \left((t-u)^{H-\frac{1}{2}} - (-u)^{H-\frac{1}{2}} \right) dB_u + \int_0^t (t-u)^{H-\frac{1}{2}} dB_u \right) \end{aligned}$$

⁸To see this is the unique solution we have merely used graphical methods.

⁹Note that the power term in the integrand is $H + \frac{1}{2} - 1 = H - \frac{1}{2}$.

In fact the above process is actually (up to scaling) fBm_H and turns out to be one of the most important representations.

Theorem 1.3.1 (Mandelbrot-Vann Ness / Time Representation of fBm_H [16]).

$$\bar{B}_t^H = \frac{1}{c_H} \left(\int_{-\infty}^0 \left((t-u)^{H-\frac{1}{2}} - (-u)^{H-\frac{1}{2}} \right) dB_u + \int_0^t (t-u)^{H-\frac{1}{2}} dB_u \right)$$

where

$$c_H = \sqrt{\frac{1}{2H} + \int_0^\infty \left((1+u)^{H-\frac{1}{2}} - u^{H-\frac{1}{2}} \right)^2 du}$$

is a fBm_H.

Proof. It is clear that an integral with respect to a Gaussian process is again a Gaussian process, and therefore $\bar{B}^H$ is also a Gaussian process. Furthermore we can see that $\mathbb{E}(\bar{B}_t^H) = 0$. Therefore, $\bar{B}^H$ is a centred Gaussian process.

Now, to show that $\bar{B}^H$ has stationary increments, we rely on the fact that the Wiener process has stationary and independent increments. So taking some $s > t$ we have

$$\begin{aligned} (\bar{B}_t^H - \bar{B}_s^H) &= \frac{1}{c_H^2} \int_{\mathbb{R}} (t-u)_+^{H-\frac{1}{2}} - (s-u)_+^{H-\frac{1}{2}} dB_u \\ &= \frac{1}{c_H^2} \int_{\mathbb{R}} (t-s-v)_+^{H-\frac{1}{2}} - (-v)_+^{H-\frac{1}{2}} dB_{v+s} \quad (\text{where } v := u-s) \\ &\stackrel{d}{=} \frac{1}{c_H^2} \int_{\mathbb{R}} (t-s-v)_+^{H-\frac{1}{2}} - (-v)_+^{H-\frac{1}{2}} dB_v \quad (dB \text{ is stationary}) \\ &= (\bar{B}_{t-s}^H) \end{aligned}$$

Therefore, $\bar{B}^H$ has stationary increments. Now it remains to show that $\bar{B}^H$ satisfies the covariance structure from 1.2.1.

Take $t > 0$, then by substituting $v := \frac{u}{t}$ for $t > 0$ and $v := 1 - \frac{u}{t}$ for $t < 0$ and factoring out our t terms along with using Ito isometry we get

$$\begin{aligned} \mathbb{E}[(\bar{B}_t^H)^2] &= \frac{1}{c_H^2} \int_{\mathbb{R}} \left((t-u)_+^{H-\frac{1}{2}} - (-u)_+^{H-\frac{1}{2}} \right)^2 du \\ &= \frac{1}{c_H} |t|^{2H} \int_{\mathbb{R}} \left((1-v)_+^{H-\frac{1}{2}} - (-v)_+^{H-\frac{1}{2}} \right)^2 dv \quad (du = t dv) \\ &= \frac{1}{c_H} |t|^{2H} \left(\int_{-\infty}^0 \left((1-v)^{H-\frac{1}{2}} - (-v)^{H-\frac{1}{2}} \right)^2 dv + \int_0^1 (1-v)^{H-\frac{1}{2}} dv \right) \\ &= \frac{1}{c_H} |t|^{2H} \left(\int_{-\infty}^0 \left((1-v)^{H-\frac{1}{2}} - (-v)^{H-\frac{1}{2}} \right)^2 dv + \frac{1}{2H} \right) \\ &= |t|^{2H} \end{aligned}$$

We then easily recover the required covariance since

$$\begin{aligned}
& \mathbb{E}[(B_t^H - B_s^H)^2] = |t - s|^{2H} \quad (\text{Since } \bar{B} \text{ is stationary}) \\
\implies & \mathbb{E}[(B_t^H)^2] + \mathbb{E}[(B_s^H)^2] - 2\mathbb{E}[B_t^H B_s^H] = |t - s|^{2H}. \\
\implies & \text{Cov}(B_s^H, B_t^H) = \mathbb{E}[B_s^H B_t^H] = \frac{1}{2} [s^{2H} + t^{2H} - |s - t|^{2H}].
\end{aligned}$$

□

This integral representation is itself the proof of the existence of fBm_H which was promised earlier. While the Mandelbrot-Van Ness representation is valuable both as a pedagogical tool demonstrating long-range dependence (see section 1.2.2) and for proving existence, it has a fundamental limitation. Since the integral runs over $(-\infty, t)$, the process B_t^H depends on the driving Brownian motion B_u for $u \in (-\infty, 0)$, meaning B_t^H is not adapted to the natural filtration $\mathcal{F}_t = \sigma(B_s, 0 \leq s \leq t)$. However standard techniques of martingale theory and Itô calculus requires adaptedness (see [20]) of precisely such a filtration, and so cannot be directly applied to the Mandelbrot-Van Ness representation. To fix this, we instead seek a representation of the form:

$$B_t^H = \int_0^t K_H(t, s) dB_s$$

where K_H is a deterministic Volterra type kernel so that B_t^H depends only on W_s for $s \in [0, t]$ and adaptedness is recovered.

Theorem 1.3.2 (Volterra Kernel representation [16]). *Let $H \in (0, \frac{1}{2}) \cup (\frac{1}{2}, 1)$ and, for $t > s > 0$, set*

$$K_H(t, s) = \begin{cases} C_H^+ \cdot s^{\frac{1}{2}-H} \int_s^t (u-s)^{H-\frac{3}{2}} u^{H-\frac{1}{2}} du & \text{if } H > \frac{1}{2}, \\ C_H^- \cdot \left\{ \left(\frac{t}{s}\right)^{H-\frac{1}{2}} (t-s)^{H-\frac{1}{2}} - \left(H - \frac{1}{2}\right) s^{\frac{1}{2}-H} \int_s^t u^{H-\frac{3}{2}} (u-s)^{H-\frac{1}{2}} du \right\} & \text{if } H < \frac{1}{2}, \end{cases}$$

where

$$C_H^+ := \sqrt{\frac{H(2H-1)}{\int_0^1 (1-x)^{1-2H} x^{H-\frac{3}{2}} dx}}, \quad C_H^- := \sqrt{\frac{2H}{(1-2H) \int_0^1 (1-x)^{-2H} x^{H-\frac{1}{2}} dx}}.$$

Then the process

$$B_t^H = \int_0^t K_H(t, s) dW_s$$

is a fBm_H.

We choose to omit the lengthy proof of this theorem. This Volterra representation is crucial in the development of a theory of integration with respect to fBm - which will not be developed any further here (see [16]).

1.4 The Ornstein-Uhlenbeck Representation

As discussed, the non-Markovian nature of fBm makes it particularly difficult to work with, in particular Markovian process allow us to apply techniques such as Feynman-Kac as well as efficient simulation. In this section we will show how we can lift fBm_H to a Markov process - a technique known as Markovian lifting [10]. In particular we will be lifting to an infinite dimensional Ornstein-Uhlenbeck process, thus it is important that we first establish such a process and demonstrate that it is indeed Markovian.

1.4.1 Ornstein-Uhlenbeck Process

The Ornstein-Uhlenbeck (OU) process $(U_t)_{t \geq 0}$ is a mean-reverting process defined by the following stochastic differential equation:

$$dU_t := -\beta(U_t - \mu)dt + \sigma dB_t.$$

Here μ is our expected value of the process; σ^2 is the variance; and β is our mean reversion rate. For the remainder of this chapter we will be working with the standard OU process where $\mu = 0$ and $\sigma = 1$ which we will denote by $(Z_t^\beta)_{t \geq 0}$ where

$$dZ_t^\beta := -\beta Z_t^\beta dt + dB_t.$$

Now by applying Ito's lemma [7] with the function $f(t, x) = xe^{\beta t}$ it follows that

$$\begin{aligned} d(Z_t^\beta e^{\beta t}) &= e^{\beta t} dB_t \\ \implies Z_t^\beta &= \int_{-\infty}^t e^{-\beta(t-u)} dB_u \end{aligned}$$

Where we choose to integrate from $-\infty$ to ensure that the process is stationary and that no single point is over emphasised at the beginning. Notice also that Z_t^β is Gaussian with mean zero since it is a stochastic integral with deterministic kernel. Furthermore we can calculate the variance as follows:

$$\begin{aligned} \text{Var}(Z_t^\beta) &= \mathbb{E}[(Z_t^\beta)^2] = \mathbb{E} \left[\left(\int_{-\infty}^t e^{-\beta(t-u)} dB_u \right)^2 \right] \\ &= \int_{-\infty}^t e^{-2\beta(t-u)} du \quad (\text{By Ito isometry}) \\ &= \frac{1}{2\beta}. \end{aligned}$$

Thus we can simply write $Z_t^\beta \sim N(0, \frac{1}{2\beta})$. This makes sense as we think of Z_t^β as being a stationary process that is mean reverting, where the faster we are reverting to the mean (larger β) would correspond to a smaller variance of the process.

1.4.2 Multivariate Ornstein-Uhlenbeck process

We can now generalize this process to be multi-dimensional, namely for arbitrary $m \in \mathbb{N}$, let $\beta = (\beta_1, \dots, \beta_m)^T \in \mathbb{R}_+^m$, and $\mathbf{Z}_t^\beta = (Z_t^{\beta_1}, \dots, Z_t^{\beta_m})^T$. Following [15], we can describe $(\mathbf{Z}_t^\beta)_{t \geq 0}$ as a multivariate Ornstein-Uhlenbeck process with the following integral representation¹⁰:

$$\mathbf{Z}_t^\beta = \int_{-\infty}^t e^{-\text{diag}(\beta)(t-u)} I_m dB_u$$

The vector integrals are evaluated element-wise. Now, the transition densities for $(\mathbf{Z}_t^\beta)_{t \geq 0}$ will be required to show the Markov property, so we compute them here. Fix arbitrary $s, t \in \mathbb{R}_{\geq 0}$ such that $s < t$. Then

$$\begin{aligned} \mathbf{Z}_t^\beta &= \int_{-\infty}^t e^{-\text{diag}(\beta)(t-u)} \mathbf{1}_m dB_u \\ &= e^{-\text{diag}(\beta)(t-s)} \int_{-\infty}^s e^{-\text{diag}(\beta)(s-u)} I_m dB_u + \int_s^t e^{-\text{diag}(\beta)(t-u)} I_m dB_u \\ \mathbf{Z}_t^\beta &= e^{-\text{diag}(\beta)(t-s)} \mathbf{Z}_s^\beta + \int_s^t e^{-\text{diag}(\beta)(t-u)} I_m dB_u \end{aligned}$$

Notice that the second term is a multivariate Gaussian¹¹ hence the conditional distribution of $\mathbf{Z}_t^\beta \mid \mathbf{Z}_s^\beta$ is fully specified by its mean vector and covariance matrix (Since $\mathbf{Z}_s^\beta$ is a pre-determined constant vector in this case). Thus the mean is given by

$$\begin{aligned} \mu_{t|s, \mathbf{Z}_s^\beta} &:= \mathbf{E} \left[\mathbf{Z}_t^\beta \mid \mathbf{Z}_s^\beta \right] = \mathbf{E} \left[e^{-\text{diag}(\beta)(t-s)} \mathbf{Z}_s^\beta \mid \mathbf{Z}_s^\beta \right] + \mathbf{E} \left[\int_s^t e^{-\text{diag}(\beta)(t-u)} I_m dB_u \mid \mathbf{Z}_s^\beta \right] \\ &= e^{-\text{diag}(\beta)(t-s)} \mathbf{Z}_s^\beta + \mathbf{0}_m = e^{-\text{diag}(\beta)(t-s)} \mathbf{Z}_s^\beta \end{aligned}$$

and the covariance matrix by

$$\begin{aligned} \Sigma_{t|s, \mathbf{Z}_s^\beta}^\beta &= \mathbf{E} \left[\left(\mathbf{Z}_t^\beta - \mathbf{E} \left[\mathbf{Z}_t^\beta \mid \mathbf{Z}_s^\beta \right] \right) \left(\mathbf{Z}_t^\beta - \mathbf{E} \left[\mathbf{Z}_t^\beta \mid \mathbf{Z}_s^\beta \right] \right)^T \mid \mathbf{Z}_s^\beta \right] \\ &= \mathbf{E} \left[\left(\int_s^t e^{-\text{diag}(\beta)(t-u)} \mathbf{1}_m dB_u \right) \left(\int_s^t e^{-\text{diag}(\beta)(t-u)} \mathbf{1}_m dB_u \right)^T \mid \mathbf{Z}_s^\beta \right] \\ &= \left[\mathbf{E} \left[\left(\int_s^t e^{-\beta_i(t-u)} dB_u \right) \left(\int_s^t e^{-\beta_j(t-u)} dB_u \right) \right] \right]_{i,j=1}^m \\ &= \left[\int_s^t e^{-\beta_i(t-u)} e^{-\beta_j(t-u)} du \right]_{i,j=1}^m \quad (\text{By Ito isometry}) \\ \Sigma_{t|s, \mathbf{Z}_s^\beta}^\beta &= \left[\frac{1 - e^{-(\beta_i + \beta_j)(t-s)}}{\beta_i + \beta_j} \right]_{i,j=1}^m \end{aligned}$$

¹⁰Here we use the standard definition of matrix exponentiation $e^A := \sum_{n=0}^{\infty} \frac{A^n}{n!}$

¹¹The second term is a vector of Ito integrals with respect to a Brownian motion, hence is a multivariate normal.

Thus, $\mathbf{Z}_t^\beta \mid \mathbf{Z}_s^\beta \sim \mathcal{N}(\boldsymbol{\mu}_{t|s, \mathbf{Z}_s^\beta}, \boldsymbol{\Sigma}_{t|s, \mathbf{Z}_s^\beta})$ and hence has transition density:

$$f_{t|s, \mathbf{Z}_s^\beta}(\mathbf{z}) = \frac{1}{(2\pi)^{m/2} \sqrt{\det(\boldsymbol{\Sigma}_{t|s, \mathbf{Z}_s^\beta})}} \exp \left\{ -\frac{1}{2} (\mathbf{z} - \boldsymbol{\mu}_{t|s, \mathbf{Z}_s^\beta})^T \boldsymbol{\Sigma}_{t|s, \mathbf{Z}_s^\beta}^{-1} (\mathbf{z} - \boldsymbol{\mu}_{t|s, \mathbf{Z}_s^\beta}) \right\}$$

We next extend this notion to be infinite dimensional and demonstrate that it is indeed Markovian.

1.4.3 Infinite Dimensional OU process & the Markov Property

In the next section we will want to consider an OU process where the mean reversion speed can be any value in the continuum $\beta \in (0, \infty)$. Thus we must make sense of what such an infinite dimensional extension would look like¹². Thus suppose $t > 0$ is fixed and consider the infinite dimensional vector $\mathbf{Z}_t := (Z_t^\beta)_{\beta > 0}$ which we will denote by Z_t to emphasize the fact that β varies. It is demonstrated in [19] that this infinite dimensional process over all possible values of β actually exists and that we can define a sensible notion of probability distributions over such a space.

Now, to show that Z_t is Markovian, it is shown in [19] that it suffices to show the Markov property holds jointly on every finite-dimensional “cylinder”. By this we mean that for every $m \in \mathbb{N}$ and every finite grid $\boldsymbol{\beta} = (\beta_1, \dots, \beta_m)^T \in \mathbb{R}_+^m$, the vector $\mathbf{Z}_t^\beta = (Z_t^{\beta_1}, \dots, Z_t^{\beta_m})^T$ must satisfy

$$\mathbf{Z}_t^\beta \mid \mathcal{F}_s \stackrel{d}{=} \mathbf{Z}_t^\beta \mid \mathbf{Z}_s^\beta.$$

Crucially, this must hold jointly across the coordinates $\beta_1, \dots, \beta_m$ for an arbitrary finite grid - showing the property coordinate-by-coordinate for each β_i in isolation is not sufficient, since it would say nothing about the joint dependence structure between coordinates. We now fix an arbitrary $m \in \mathbb{N}$ and grid $\boldsymbol{\beta} \in \mathbb{R}_+^m$ and prove exactly this joint statement, drawing on the multivariate OU process established above.

Since random variables are uniquely determined by their characteristic functions [7], it suffices to show that the corresponding characteristic functions agree,

¹²In this section we are non-rigorous in defining such an infinite dimensional process as we want to stick to the main idea of this project so as to not get lost in the measure theoretic details (see [19]).

as follows.

$$\begin{aligned}
\mathbf{E}\left[e^{i\mathbf{x}'\mathbf{Z}_t^\beta} \mid \mathcal{F}_s\right] &= \mathbf{E}\left[\exp\left\{i\mathbf{x}'\left(e^{-\text{diag}(\beta)(t-s)}\mathbf{Z}_s^\beta + \int_s^t e^{-\text{diag}(\beta)(t-u)}\mathbf{1}_m dB_u\right)\right\} \mid \mathcal{F}_s\right] \\
&= \exp\left\{i\mathbf{x}'\left(e^{-\text{diag}(\beta)(t-s)}\mathbf{Z}_s^\beta\right)\right\} \mathbf{E}\left[\exp\left\{i\mathbf{x}'\left(\int_s^t e^{-\text{diag}(\beta)(t-u)}\mathbf{1}_m dB_u\right)\right\} \mid \mathcal{F}_s\right] \\
&= \exp\left\{i\mathbf{x}'\left(e^{-\text{diag}(\beta)(t-s)}\mathbf{Z}_s^\beta\right)\right\} \mathbf{E}\left[\exp\left\{i\mathbf{x}'\left(\int_s^t e^{-\text{diag}(\beta)(t-u)}\mathbf{1}_m dB_u\right)\right\} \mid \mathbf{Z}_s^\beta\right] \\
&= \mathbf{E}\left[\exp\left\{i\mathbf{x}'\left(e^{-\text{diag}(\beta)(t-s)}\mathbf{Z}_s^\beta + \int_s^t e^{-\text{diag}(\beta)(t-u)}\mathbf{1}_m dB_u\right)\right\} \mid \mathbf{Z}_s^\beta\right] \\
&= \mathbf{E}\left[e^{i\mathbf{x}'\mathbf{Z}_t^\beta} \mid \mathbf{Z}_s^\beta\right].
\end{aligned}$$

For the third equality, we used the fact that the random variable in the expectation is independent of both $\mathcal{F}_s$ and $\mathbf{Z}_s^\beta$. Characteristic functions completely specify their corresponding distributions, so $\mathbf{Z}_t^\beta \mid \mathcal{F}_s \stackrel{d}{=} \mathbf{Z}_t^\beta \mid \mathbf{Z}_s^\beta$ as required.

Since $m \in \mathbb{N}$ and the grid $\beta \in \mathbb{R}_+^m$ were arbitrary, the above shows that the Markov property holds jointly on every finite-dimensional cylinder. Hence, by [19], the infinite dimensional Z_t is itself Markovian, as required.

1.4.4 Derivation of Muravlev Representation

Now that we have built up the required understanding of an Ornstein-Uhlenbeck process and extended it to infinitely many dimensions, we are ready to fully appreciate how it can be used to represent fBm_H . The derivation essentially comes down to algebraic trickery using the Gamma function, however a full appreciation of the importance requires our developed knowledge of the OU process from the last section. The next theorem is the key algebraic trick that allows us to develop the so called Muravlev representation.

Theorem 1.4.1 (Ornstein-Uhlenbeck representation of the fractional kernel). *For $t \in \mathbb{R}_{>0}$ and $H \in (0, 1/2) \cup (1/2, 1)$ we have the following formulae.*

$$K(t) = \frac{t^{H-1/2}}{\Gamma(\frac{1}{2} - H)} = \begin{cases} \frac{1}{\Gamma(\frac{1}{2} - H)} \int_0^\infty \beta^{-1/2-H} e^{-t\beta} d\beta, & H \in (0, 1/2) \\ \frac{1}{\Gamma(\frac{1}{2} - H)} \int_0^\infty \beta^{-1/2-H} (e^{-t\beta} - 1) d\beta, & H \in (1/2, 1) \end{cases}$$

Proof. We present here a sketch of that given by [19]. We begin with the easier case $H \in (0, 1/2)$, which simply follows from the known Laplace transform identity in which $t \mapsto t^{H-1/2}$ is the Laplace transform of the function $\beta \mapsto \beta^{-1/2-H}/\Gamma(1/2 - H)$. This can also easily be shown from scratch using the substitution $u = \beta t$ which we do not present here.

Now the $H \in (1/2, 1)$ case is a bit more intricate. In fact the integral definition of $\Gamma(x)$ does not converge for $x < 0$, however (noting that we want

this to extend the factorial function) the definition can be extended via the identity $x\Gamma(x) = \Gamma(x+1)$. In particular, for $x \in (-1, 0)$ one has

$$\begin{aligned}\Gamma(x) &= \frac{1}{x}\Gamma(x+1) = \frac{1}{x} \int_0^\infty u^x e^{-u} du \\ &= \frac{1}{x} \left([-u^x (e^{-u} - 1)]_{u=0}^\infty + \int_0^\infty x u^{x-1} (e^{-u} - 1) du \right) = \int_0^\infty u^{x-1} (e^{-u} - 1) du\end{aligned}$$

using integration by parts. Then substituting $u = \beta t$ into the integral on the right-hand side of gives we see that

$$\begin{aligned}\int_0^\infty \beta^{-1/2-H} (e^{-\beta t} - 1) d\beta &= t^{H-1/2} \int_0^\infty u^{(1/2-H)-1} (e^{-u} - 1) du \\ &= t^{H-1/2} \Gamma(1/2 - H)\end{aligned}$$

□

We now derive the Muravlev preservation by simply applying this trick to the Mandelbrot-Vann Ness representation.

Theorem 1.4.2 (Muravlev representation). *Let $H \in (0, 1/2) \cup (1/2, 1)$. For each $\beta \in \mathbb{R}_{>0}$, let $(Z_t^\beta)_{t \geq 0}$ be the standard Ornstein-Uhlenbeck process. If*

$$\bar{B}_t^H = \begin{cases} \frac{1}{\bar{c}_H} \int_0^\infty \beta^{-1/2-H} \left(Z_t^\beta - Z_0^\beta \right) d\beta, & \text{if } H \in (0, 1/2) \\ \frac{1}{\bar{c}_H} \int_0^\infty \beta^{-1/2-H} \left(Z_t^\beta - Z_0^\beta - B_t \right) d\beta, & \text{if } H \in (1/2, 1), \end{cases}$$

where the constant

$$\bar{c}_H = c_H \times \Gamma\left(\frac{1}{2} - H\right) = \Gamma\left(\frac{1}{2} - H\right) \sqrt{\frac{1}{2H} + \int_0^\infty \left((1+u)^{H-\frac{1}{2}} - u^{H-\frac{1}{2}} \right)^2 du}$$

then $(\bar{B}_t^H)_{t \in \mathbb{R}_{\geq 0}}$ is a fBm_H .

Proof. We start with the Mandelbrot-Van Ness representation. For $t \in \mathbb{R}_{\geq 0}$,

$$\begin{aligned}B_t^H &= \frac{1}{c_H} \int_{\mathbb{R}} \left((t-s)_+^{H-1/2} - (-s)_+^{H-1/2} \right) dB_s. \\ &= \frac{1}{c_H} \int_{-\infty}^0 \left((t-s)^{H-1/2} - (-s)^{H-1/2} \right) dB_s + \frac{1}{c_H} \int_0^t (t-s)^{H-1/2} dB_s.\end{aligned}$$

Applying Theorem 1.4.1 to each power $(t-s)^{H-1/2}$ and $(-s)^{H-1/2}$, and noting that $\bar{c}_H = c_H \Gamma(1/2 - H)$. that gives us the first case $H \in (0, \frac{1}{2})$

$$\begin{aligned}B_t^H &= \frac{1}{\bar{c}_H} \int_{-\infty}^0 \left[\int_0^\infty \beta^{-1/2-H} \left(e^{-\beta(t-s)} - e^{\beta s} \right) d\beta \right] dB_s \\ &\quad + \frac{1}{\bar{c}_H} \int_0^t \left[\int_0^\infty \beta^{-1/2-H} e^{-\beta(t-s)} d\beta \right] dB_s.\end{aligned}$$

and for the case when $H \in (1/2, 1)$:

$$B_t^H = \frac{1}{\bar{c}_H} \int_{-\infty}^0 \left[\int_0^\infty \beta^{-1/2-H} \left(e^{-\beta(t-s)} - e^{\beta s} \right) d\beta \right] dB_s \\ + \frac{1}{\bar{c}_H} \int_0^t \left[\int_0^\infty \beta^{-1/2-H} \left(e^{-\beta(t-s)} - 1 \right) d\beta \right] dB_s.$$

Now using a stochastic Fubini theorem (see [7]) to interchange the order of integration and recalling,

$$Z_t^\beta := \int_{-\infty}^t e^{-\beta(t-s)} dB_s$$

we get

$$B_t^H = \begin{cases} \frac{1}{\bar{c}_H} \int_0^\infty \beta^{-1/2-H} \left(Z_t^\beta - Z_0^\beta \right) d\beta, & \text{if } H \in (0, 1/2), \\ \frac{1}{\bar{c}_H} \int_0^\infty \beta^{-1/2-H} \left(Z_t^\beta - Z_0^\beta - B_t \right) d\beta, & \text{if } H \in (1/2, 1), \end{cases}$$

which is Muravlev's OU representation. $\square$

This representation is much easier to work with as it allows us to instead work with Z_t^β which we established is Markovian. Moreover it provides Heuristic insights as to the inner workings of fBm_H as we can think of it as simply a linear combination of OU process' with different mean reversion. We next use this representation to approximate fBm_H .

1.5 Simulating fBm

Now that we have seen all of the basic theory, the natural next step is the implementation of fBm_H simulation. The fact that fBm_H is non-Markovian makes simulation methods difficult both in terms of the mathematics and computational cost. In this section we will present the Cholesky method for simulating fBm_H where we essentially just simulate the covariance between all increments. In section 2 we will demonstrate another simulation method that makes use of both the Cholesky method and also an integral representation.

1.5.1 Cholesky Method

The Cholesky method is a standard tool for simulating Gaussian processes. $A \in \mathbb{R}^{m \times n}$ is a matrix, $\mu \in \mathbb{R}^n$ a vector then we can define $\mathbf{X} = \mu + A\mathbf{Z}$. Then basic manipulation shows that $\mathbf{X}$ has a multivariate Gaussian distribution with covariance matrix $\Sigma = AA^T$ and mean $\mu = \mathbb{E}[\mathbf{X}]$.

Note that any positive semi-definite matrix Σ has a Cholesky decomposition $\Sigma = LL^T$, where L is a lower triangular matrix. Thus following the fact that all covariance matrices are positive semi-definite, we therefore can formulate the following algorithm for simulating a multivariate Gaussian.

Algorithm 1

Input:

- $\Sigma \in \mathbb{R}^{n \times n}$ positive definite covariance matrix
- $\mu \in \mathbb{R}^n$ mean vector
- $m \in \mathbb{N}$ number of samples

Output:

- m samples of multivariate Gaussian with mean μ and covariance Σ .

1. Compute L such that $\Sigma = LL^T$.
 2. Repeat the following m times:
 - a) Simulate n independent samples from the standard normal distribution $\mathbf{Z} = (Z_1, \dots, Z_n)$.
 - b) Save new sample $\mu + L\mathbf{Z}$.
-

To implement this algorithm we further need to understand how to proceed with steps 1. and 2.a). The simulation of independent normal random variables is beyond the scope of this report¹³. Regarding the computation of the Cholesky decomposition, one can use the fact that L is lowertriangular to compute its entries iteratively from the top to the bottom. Here is a simple Python code that works for positive definite matrices:

```
1 import numpy as np
2
3 def cholesky(A):
4     L = np.zeros_like(A)
5     n, _ = A.shape
6     for i in range(n):
7         for j in range(i + 1):
8             sum_ = np.sum(L[i] * L[j])
9             if i == j:
10                 L[i, j] = np.sqrt(A[i, i] - sum_)
11             else:
12                 L[i, j] = (A[i, j] - sum_) / L[j, j]
13     return L
```

Note that the computational cost of this algorithm is of order¹⁴

We now consider fBm_H which is a centred Gaussian process with $C(t, s) = Cov(X_t, X_s) = \frac{1}{2}[t^{2H} + s^{2H} - |t - s|^{2H}]$. Hence if we use n time steps over a time intervals T we can discretize on a time grid $\{0 = t_0 < t_1 < \dots < t_n = T\}$ which then simply amounts to the simulation of the $(n + 1)$ -dimensional Gaussian random variable $\mathbf{X} := (X_{t_0}, \dots, X_{t_n})$ which we can write as $\mathbf{X} = \Sigma \mathbf{Z}$ where $(\Sigma)_{ij} = C(t_i, t_j)$, we then simply apply the Cholesky algorithm¹⁵

¹³In Python we can use the numpy package and the command `numpy.random.randn(n)`.

¹⁴Notice that a third inner loop is hidden in the numpy sum)

¹⁵A full Python implementation can be found at [5] to simulate fBm_H using the Cholesky algorithm.

Chapter 2

Lifted Fractional Brownian Motion

We now leverage the Muravlev representation to construct a genuinely Markovian, finite-dimensional approximation of fBm_H – the central original contribution of this report. A Markovian approximation is valuable precisely because it restores access to standard techniques that rely on the Markov property, such as Feynman-Kac arguments, which we cannot apply to fBm_H directly.

We call this construction *lifted fBm*, in direct analogy with the lifted Heston model [10], whose node and weight scheme we adapt.

In this chapter we adapt the lifted Heston node scheme to the fBm kernel and extend it to stationary, two-sided fBm (Section 2.1). We then prove L^2 convergence for $H \in (0, 1/2)$ (Section 2.2), and extend the construction, with numerical support in place of a full proof, to $H \in (1/2, 1)$ (Section 2.3).

2.1 Discretisation Scheme

Recall from Theorem 1.4.2 that for $H \in (0, 1/2)$, the fractional Brownian motion admits the Markovian OU-superposition representation

$$B_t^H = \frac{1}{\bar{c}_H} \int_0^\infty \beta^{-1/2-H} (Z_t^\beta - Z_0^\beta) d\beta, \quad Z_t^\beta = \int_{-\infty}^t e^{-\beta(t-s)} dB_s. \quad (2.1)$$

We initially attempted to discretise (2.1) using a naive geometric grid $\beta_j = r^j$, $j = -n + 1, \dots, n$, for fixed $r \in (1, 2)$, with weights obtained by directly integrating the kernel over each subinterval. This proved ineffective: for a fixed r , the grid cannot simultaneously resolve both the slow (long-memory) and fast (short-timescale) modes with a modest number of nodes widening the range in one direction starves resolution in the other. Since this failure mode is a property of the quadrature itself rather than of the stochastic integral, we can test it directly on the deterministic fractional kernel $K_H(t-u)$ before extending

to the full stochastic construction. Figure 2.1 shows this comparison directly. This suggests that a Markovian approximation of fBm_H is a non-trivial problem

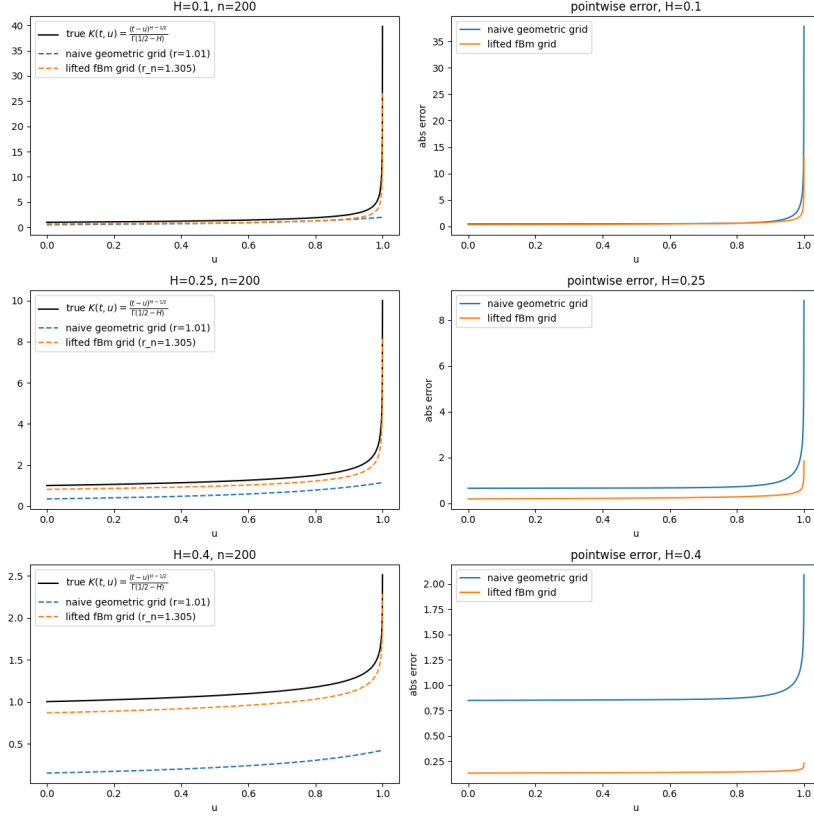


Figure 2.1: We use the exact same quadrature scheme to construct a finite approximation of the fractional kernel $K_H(t-u) = \frac{(t-u)^{H-\frac{1}{2}}}{\Gamma(\frac{1}{2}-H)}$ for $t=1$, which empirically demonstrates that our lifted fBm quadrature is much more accurate, particularly for larger H .

We instead use the same node and weight parametrisation underlying the lifted Heston approximation (our theorem 2.2.1, originally from [10]), applied directly to the fBm kernel itself rather than to the rough Heston variance kernel which will be discussed further in 3.2. We will first state the discretisation before deriving its origin. For $\alpha = H + \frac{1}{2}$ and a sequence $r_n \downarrow 1$ with $n \log(r_n) \rightarrow \infty$,

we take

$$w_i^n = \frac{(r_n^{1-\alpha} - 1) r_n^{(\alpha-1)(1+n/2)}}{\Gamma(\alpha)\Gamma(2-\alpha)} r_n^{(1-\alpha)i}, \quad (2.2)$$

$$\beta_i^n = \left(\frac{1-\alpha}{2-\alpha} \right) \frac{r_n^{2-\alpha} - 1}{r_n^{1-\alpha} - 1} r_n^{i-1-n/2}, \quad (2.3)$$

for $i = 1, \dots, n$ giving the finite Markovian approximation which we will hereafter refer to as *lifted fBm_H*

$$B_t^{H,n} = \sum_{i=1}^n w_i^n (Z_t^{\beta_i^n} - Z_0^{\beta_i^n}). \quad (2.4)$$

In [10] the schedule used is $r_n = 1 + \frac{10}{n^{0.9}}$ however we found that the practical schedule $r_n = \exp(1/n^\gamma)$ for, $0 < \gamma < 1$, which satisfies $r_n \downarrow 1$ and $n \log(r_n) = n^{1-\gamma} \rightarrow \infty$ as required. In particular we found $\gamma = 0.25$ worked well in practice. Unlike our initial fixed- r attempt, this parametrization concentrates node density correctly across the full range of relevant timescales as n grows, and reproduces the fractional kernel $(t-u)^{H-1/2}$ substantially more accurately for a given number of nodes.

Each factor $Z^{\beta_i^n}$ is simulated using its exact stationary transition law (Appendix A with the joint initial vector $(Z_0^{\beta_1^n}, \dots, Z_0^{\beta_n^n})$ drawn via a Cholesky decomposition of $\Gamma_{i,j} = 1/(\beta_i^n + \beta_j^n)$, and all factors sharing the same driving Brownian increments at each time step.

2.1.1 Origin of the node and weight formulas

The nodes and weights (β_i^n, w_i^n) arise from partitioning $(0, \infty)$ into geometric blocks

$$\eta_i^n := r_n^{i-n/2}, \quad i = 0, \dots, n,$$

and representing the measure $\mu(dx) = x^{-\alpha}/(\Gamma(\alpha)\Gamma(1-\alpha)) dx$ restricted to each block $[\eta_{i-1}^n, \eta_i^n]$ by a single point mass $w_i^n \delta_{\beta_i^n}$, chosen to match the block's weight and centre of mass under μ :

$$w_i^n := \mu([\eta_{i-1}^n, \eta_i^n]) = \int_{\eta_{i-1}^n}^{\eta_i^n} \mu(dx), \quad \beta_i^n := \frac{1}{w_i^n} \int_{\eta_{i-1}^n}^{\eta_i^n} x \mu(dx).$$

That is, w_i^n is exactly the μ -mass of the i -th block, and β_i^n is the μ -weighted mean (centroid) of x within that block - the single point which reproduces the block's contribution to $\int x \mu(dx)$ exactly, rather than an arbitrary representative point such as the grid boundary η_i^n itself. It is this identity, in particular $w_i^n = \mu([\eta_{i-1}^n, \eta_i^n])$, that is used in the proof of Theorem 2.2.1 below.

Evaluating both integrals in closed form using $\Gamma(2-\alpha) = (1-\alpha)\Gamma(1-\alpha)$ and substituting $\eta_i^n = r_n^{i-n/2}$ yields precisely the formulas of equation (2.6):

$$w_i^n = \frac{(r_n^{1-\alpha} - 1) r_n^{(\alpha-1)(1+n/2)}}{\Gamma(\alpha)\Gamma(2-\alpha)} r_n^{(1-\alpha)i}, \quad \beta_i^n = \left(\frac{1-\alpha}{2-\alpha} \right) \frac{r_n^{2-\alpha} - 1}{r_n^{1-\alpha} - 1} r_n^{i-1-n/2}.$$

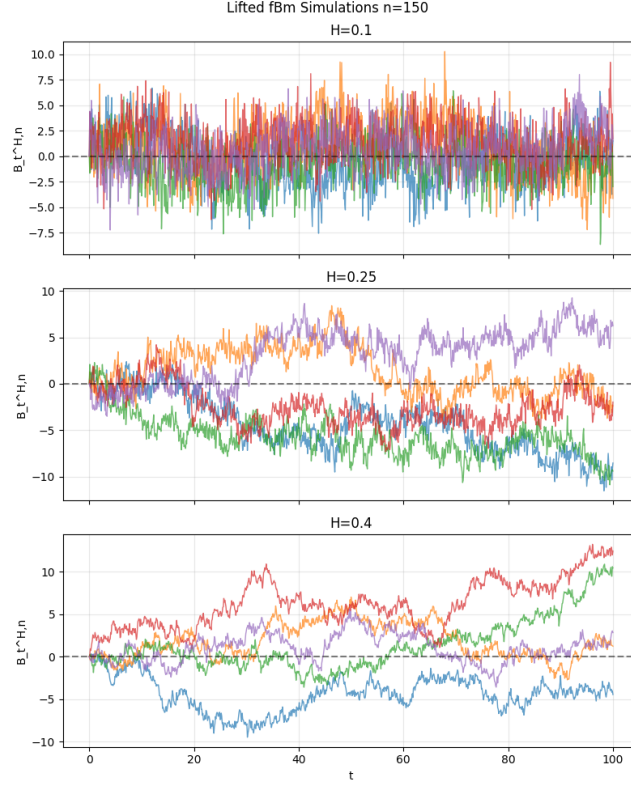


Figure 2.2: Sample paths of lifted fractional Brownian motion for $H = 0.1, 0.25, 0.4$. Here $n = 150$ and $r_n = \exp(n^\gamma) = \exp(1/150^{0.25}) \approx 1.331$. Implementation available at [5].

Proof. Writing $\alpha = H + \frac{1}{2}$, the integral yield the weight over each block is

$$w_i^n = \int_{\eta_{i-1}^n}^{\eta_i^n} \frac{x^{-\alpha}}{\Gamma(\alpha)\Gamma(1-\alpha)} dx = \frac{(\eta_i^n)^{1-\alpha} - (\eta_{i-1}^n)^{1-\alpha}}{\Gamma(\alpha)\Gamma(2-\alpha)},$$

using $\Gamma(2-\alpha) = (1-\alpha)\Gamma(1-\alpha)$. Substituting $\eta_i^n = r_n^{i-n/2}$ and factoring $(\eta_i^n)^{1-\alpha} - (\eta_{i-1}^n)^{1-\alpha} = r_n^{(1-\alpha)(i-1-n/2)}(r_n^{1-\alpha} - 1)$ gives the stated form of w_i^n .

Similarly, the centre of mass integral is

$$\int_{\eta_{i-1}^n}^{\eta_i^n} x \mu(dx) = \frac{1-\alpha}{(2-\alpha)\Gamma(\alpha)\Gamma(2-\alpha)} \left[(\eta_i^n)^{2-\alpha} - (\eta_{i-1}^n)^{2-\alpha} \right].$$

Dividing by w_i^n , the common factor $r_n^{(1-\alpha)(i-1-n/2)}$ cancels, leaving a single power of $\eta_{i-1}^n = r_n^{i-1-n/2}$ and yielding the stated form of β_i^n . $\square$

2.2 Convergence proof for $H < \frac{1}{2}$

We prove here that $B_t^{H,n} \rightarrow B_t^H$ in the L^2 norm. To do so we will prove that an equivalent approximation of the fractional kernel

$$K_H(t) = \frac{t^{H-\frac{1}{2}}}{\Gamma(\frac{1}{2}-H)} = \frac{1}{\Gamma(\frac{1}{2}-H)} \int_0^\infty \beta^{-1/2-H} e^{-t\beta} d\beta$$

using the same lifted fBm weights such that

$$K_H^n(t) = \sum_{i=0}^n w_i e^{-\beta_i t}$$

converges in the L^2 norm.

Theorem 2.2.1 (Kernel Convergence in L^2). *Suppose $\alpha = H + \frac{1}{2} \in (\frac{1}{2}, 1)$ and let $(r_n)_{n \geq 1}$ be a sequence of real numbers such that*

$$r_n \downarrow 1 \quad \text{and} \quad n \log(r_n) \rightarrow +\infty \quad \text{as } n \rightarrow +\infty. \quad (2.5)$$

Define $(w_i^n, \beta_i^n)_{1 \leq i \leq n}$ by

$$w_i^n = \frac{(r_n^{1-\alpha} - 1) r_n^{(\alpha-1)(1+n/2)}}{\Gamma(\alpha)\Gamma(2-\alpha)} r_n^{(1-\alpha)i}, \quad (2.6)$$

$$\beta_i^n = \left(\frac{1-\alpha}{2-\alpha} \right) \frac{r_n^{2-\alpha} - 1}{r_n^{1-\alpha} - 1} r_n^{i-1-n/2}. \quad (2.7)$$

Then for all $T > 0$,

$$\|K^n - K\|_{L^2(0,T)} \rightarrow 0 \quad \text{as } n \rightarrow +\infty, \quad (2.8)$$

where $K(t) = \int_0^{+\infty} e^{-\beta t} \mu(d\beta)$ and $\mu(d\beta) = \frac{\beta^{-H-\frac{1}{2}} d\beta}{\Gamma(H+\frac{1}{2})\Gamma(-H+\frac{1}{2})}$.

The proof given below is an expanded version of that given in [1].

Proof. The starting point is the observation, already established in Remark 2.1.1, that the weights are exactly the μ -mass of their block: setting $\eta_i^n := r_n^{i-n/2}$ for $i = 0, \dots, n$, we have $w_i^n = \mu([\eta_{i-1}^n, \eta_i^n])$ by construction. This lets us write both K and K^n as sums of the same block integrals:

$$\begin{aligned} K(t) &= \int_0^{\eta_0^n} e^{-xt} \mu(dx) + \sum_{i=1}^n \int_{\eta_{i-1}^n}^{\eta_i^n} e^{-xt} \mu(dx) + \int_{\eta_n^n}^\infty e^{-xt} \mu(dx), \\ K^n(t) &= \sum_{i=1}^n \int_{\eta_{i-1}^n}^{\eta_i^n} e^{-\beta_i^n t} \mu(dx). \end{aligned}$$

Subtracting term by term and collecting the three types of contribution - mass below the grid, the block-by-block quadrature error within the grid, and mass above the grid - gives

$$K^n(t) - K(t) = I_1(t) + I_2(t) + I_3(t), \quad (2.9)$$

where

$$\begin{aligned} I_1(t) &= \int_0^{\eta_0^n} e^{-xt} \mu(dx), \\ I_2(t) &= \sum_{i=1}^n \int_{\eta_{i-1}^n}^{\eta_i^n} (e^{-xt} - e^{-\beta_i^n t}) \mu(dx), \\ I_3(t) &= \int_{\eta_n^n}^{\infty} e^{-xt} \mu(dx). \end{aligned}$$

We bound $\|I_j\|_{L^2(0,T)}$ for each j in turn, then combine via the triangle inequality.

We begin with the mid range term I_2 . This is the hardest estimate. It measures how well a single exponential $e^{-\beta_i^n t}$, evaluated at the block's representative point, approximates e^{-xt} across the whole block $x \in [\eta_{i-1}^n, \eta_i^n]$. Since a Taylor expansion of e^{-xt} about $x = \beta_i^n$ has a remainder of order $(x - \beta_i^n)^2$, and the block width itself scales like $\eta_{i-1}^n(r_n - 1)$, we expect an error of order $(r_n - 1)^2$ per block - with an additional factor of t^2 appearing because each derivative of e^{-xt} in x brings down a factor of t . This intuition is made precise in Proposition 7.1 of [4], which we cite rather than re-derive for $t \leq T$ and each $i = 1, \dots, n$,

$$\left| \int_{\eta_{i-1}^n}^{\eta_i^n} (e^{-xt} - e^{-\beta_i^n t}) \mu(dx) \right| \leq C t^2 r_n^{1/2} (r_n - 1)^2 \int_{\eta_{i-1}^n}^{\eta_i^n} (1 \wedge x^{-1/2}) \mu(dx),$$

for a constant $C > 0$ independent of n, i, t . Summing over i and bounding the (nonnegative) sum of block integrals by the integral over the full range gives the pointwise-in- t bound

$$|I_2(t)| \leq C t^2 r_n^{1/2} (r_n - 1)^2 M, \quad M := \int_0^\infty (1 \wedge x^{-1/2}) \mu(dx) < \infty. \quad (2.10)$$

To conclude convergence in $L^2(0, T)$ we integrate the square of this bound over $t \in (0, T)$:

$$\|I_2\|_{L^2(0,T)}^2 = \int_0^T I_2(t)^2 dt \leq C^2 r_n (r_n - 1)^4 M^2 \int_0^T t^4 dt = \frac{C^2 M^2 T^5}{5} r_n (r_n - 1)^4,$$

so that

$$\|I_2\|_{L^2(0,T)} \leq C' r_n^{1/2} (r_n - 1)^2, \quad C' := C M \sqrt{T^5/5}. \quad (2.11)$$

Since $r_n \downarrow 1$ under hypothesis (2.5), the factor $(r_n - 1)^2 \rightarrow 0$ and $r_n^{1/2} \rightarrow 1$ stays bounded, so $\|I_2\|_{L^2(0,T)} \rightarrow 0$ as $n \rightarrow \infty$, for every fixed T .

We next look at the low-frequency term I_1 . Here $e^{-xt} \leq 1$ trivially gives the bound

$$|I_1(t)| \leq \int_0^{\eta_0^n} \mu(dx) = \frac{(\eta_0^n)^{1-\alpha}}{\Gamma(\alpha)\Gamma(2-\alpha)}, \quad (2.12)$$

which does not depend on t at all - it bounds I_1 uniformly over all $t \geq 0$, not just $t \in [0, T]$. Integrating the square of this constant bound over $t \in (0, T)$ therefore introduces only a factor of $\sqrt{T}$:

$$\|I_1\|_{L^2(0,T)} \leq \sqrt{T} \cdot \frac{(\eta_0^n)^{1-\alpha}}{\Gamma(\alpha)\Gamma(2-\alpha)} = \sqrt{T} \cdot \frac{r_n^{-(1-\alpha)n/2}}{\Gamma(\alpha)\Gamma(2-\alpha)} \rightarrow 0 \quad \text{as } n \rightarrow \infty, \quad (2.13)$$

using $r_n^{-(1-\alpha)n/2} \rightarrow 0$, which follows from $n \log(r_n) \rightarrow \infty$ (so that $(1-\alpha)n \log(r_n)/2 \rightarrow \infty$ too, since $\alpha < 1$) together with $r_n \rightarrow 1$.

We lastly consider the high-frequency term I_3 . This term needs a genuine $L^2(0, T)$ treatment, since $e^{-xt} \leq 1$ is too crude here to give a useful bound (the tail $\int_{\eta_n^n}^\infty \mu(dx)$ alone need not vanish quickly enough). Instead we apply Minkowski's integral inequality - the continuous analogue of the triangle inequality, moving the $L_t^2(0, T)$ -norm inside the x -integral - directly to $I_3(t) = \int_{\eta_n^n}^\infty e^{-xt} \mu(dx)$:

$$\|I_3\|_{L^2(0,T)} \leq \int_{\eta_n^n}^\infty \|e^{-x\cdot}\|_{L_t^2(0,T)} \mu(dx) = \int_{\eta_n^n}^\infty \sqrt{\frac{1-e^{-2xT}}{2x}} \mu(dx).$$

Using the elementary inequality $1 - e^{-2xT} \leq 1$,

$$\|I_3\|_{L^2(0,T)} \leq \int_{\eta_n^n}^\infty \frac{1}{\sqrt{2x}} \mu(dx) = C'' (\eta_n^n)^{1/2-\alpha} = C'' r_n^{(1/2-\alpha)n/2}, \quad (2.14)$$

for an explicit finite constant C'' depending only on α .¹ Since $\alpha \in (\frac{1}{2}, 1)$, the exponent $\frac{1}{2} - \alpha$ is strictly negative, and $\eta_n^n = r_n^{n/2} \rightarrow \infty$ by the same reasoning as above, so $\|I_3\|_{L^2(0,T)} \rightarrow 0$ as $n \rightarrow \infty$.

Combining (2.9) with the three bounds (2.11), (2.13), and (2.14) via the triangle inequality in $L^2(0, T)$,

$$\|K^n - K\|_{L^2(0,T)} \leq \|I_1\|_{L^2(0,T)} + \|I_2\|_{L^2(0,T)} + \|I_3\|_{L^2(0,T)} \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

for every fixed $T > 0$, as required. $\square$

We can use the fact that $K_H^n \xrightarrow{L^2} K_H$ from the above theorem 2.2.1 to prove the following key result of this report

¹We state C'' as a general constant here rather than the specific calculation appearing in [1], as we were unable to independently yield that constant with our own computation of the integral

Theorem 2.2.2 (Lifted fBm convergence in L^2 for $H < \frac{1}{2}$). *The lifted fBm as defined in 2.4 converges to true fBm in the L^2 sense. Formally we have that*

$$\mathbb{E}[(B_t^H - B_t^{H,n})^2] \rightarrow 0, \quad \text{as } n \rightarrow \infty.$$

Proof. Begin by recalling that for fixed $t \geq 0$

$$\begin{aligned} K_H^n(t-u) &= \sum_{i=0}^n w_i e^{-\beta_i(t-u)} \\ \Rightarrow \int_{-\infty}^t K_H^n(t-u) dB_u &= \sum_{i=0}^n \int_{-\infty}^t w_i e^{-\beta_i(t-u)} \\ &= \sum_{i=0}^n w_i Z_t^{\beta_i} \quad (\text{By the definition of } Z_t^\beta \text{ from 1.4.1}) \end{aligned}$$

and using the same argument for $t = 0$ we have that

$$\int_{-\infty}^0 K_H^n(-u) dB_u = \sum_{i=0}^n w_i Z_0^{\beta_i}.$$

Hence taking the difference yields

$$\int_{-\infty}^t K^n(t-u) dB_u - \int_{-\infty}^0 K^n(-u) dB_u = \sum_{i=0}^n w_i (Z_t^{\beta_i} - Z_0^{\beta_i})$$

Notice that the left right hand side is precisely $B_t^{H,n}$.

$$B_t^{H,n} = \int_{\mathbb{R}} K^n(t-u) \mathbf{1}_{\{u < t\}} - K^n(-u) \mathbf{1}_{\{u < 0\}} dB_u$$

So if we define $\Delta^n(t) := K^n(t) - K(t)$ then we have

$$\begin{aligned} E[(B_t^{H,n} - B_t^H)^2] &= E \left[\left(\int_{\mathbb{R}} \Delta^n(t-u) \mathbf{1}_{\{u < t\}} - \Delta^n(-u) \mathbf{1}_{\{u < 0\}} dB_u \right)^2 \right] \\ &= E \left[\left(\int_0^t \Delta^n(t-u) dB_u + \int_{-\infty}^0 (\Delta^n(t-u) - \Delta^n(-u))^2 dB_u \right)^2 \right] \\ &= \int_0^t \Delta^n(t-u)^2 du + \int_{-\infty}^0 (\Delta^n(t-u) - \Delta^n(-u))^2 du \\ &= A_n(t) + R_n(t) \end{aligned}$$

The second last line follows from Ito isometry and noting that the cross term vanishes since they are over non-intersecting regions of integration. Notice that the first term $A_n(t)$ is precisely $\|K^n(t-u) - K(t-u)\|_{L^2(0,t)}^2$ which we have established in theorem 2.2.1 goes to 0 as $n \rightarrow \infty$. Thus it remains to demonstrate that the remaining right hand integral $R_n(t)$ vanishes which we point the reader to C as the proof is long but necessary nonetheless.

□

2.3 Extension to $H > 1/2$

For $H \in (1/2, 1)$ we instead use the alternative Markovian OU-superposition representation

$$B_t^H = \bar{c}_H \int_0^\infty \beta^{-1/2-H} (Z_t^\beta - Z_0^\beta - B_t) d\beta, \quad (2.15)$$

Discretizing (2.15) with the same nodes and weights (2.3) gives our lifted fBm_H for $H > \frac{1}{2}$:

$$B_t^{H,n} = \sum_{i=1}^n w_i^n (Z_t^{\beta_i^n} - Z_0^{\beta_i^n} - B_t). \quad (2.16)$$

Numerically, this requires tracking the driving Brownian motion B_t alongside every factor $Z_t^{\beta_i^n}$, using the same shared noise increments at each step (so that B_t 's own transition remains exact, introducing no additional discretization error), and evaluating the corrected sum (2.16) directly rather than as a difference of two separately large quantities, since w_i^n can be large for small β_i^n while $(Z_t^{\beta_i^n} - Z_0^{\beta_i^n} - B_t)$ is correspondingly small - computing the difference before contracting against the weights avoids catastrophic cancellation. Figure 2.3 shows simulated sample paths using this construction, and Section 2.3.1 validates it against the true fBm covariance.

We believe an analogue of Theorem 2.2.1 holds for this representation, but proving it is beyond the scope of this report. The problem is not just technical bookkeeping: for $\alpha = H + \frac{1}{2} > 1$, the measure $\mu(dx) = x^{-\alpha} dx / (\Gamma(\alpha)\Gamma(1-\alpha))$ is no longer locally finite² near the origin, so the bounds on I_1 and I_2 underlying Theorem 2.2.1 - which rely on μ having finite mass near $x = 0$ - fail outright rather than merely requiring sharper constants. A correct proof would need to exploit the extra cancellation introduced by the $-B_t$ term to compensate for this stronger singularity, giving genuinely different convergence rates rather than a direct extension of the existing estimates. We leave this as an open direction for future work, with the numerical evidence of Section 2.3.1 suggesting the result is true.

2.3.1 Evidence Towards the $H > \frac{1}{2}$ Lifted fBm

Since no convergence proof is available to our knowledge for this regime, we instead validate (2.16) directly against the known fBm covariance $\text{Cov}(B_t^H, B_s^H) =$

²A measure μ is locally finite near the origin if $\mu((0, \epsilon)) < \infty$ for small $\epsilon > 0$. For $\mu(dx) = x^{-\alpha} dx / (\Gamma(\alpha)\Gamma(1-\alpha))$ this holds iff $\alpha < 1$ (by the p -test), which is exactly when $H < 1/2$. For $H > 1/2$ every neighbourhood of the origin already carries infinite μ -mass.

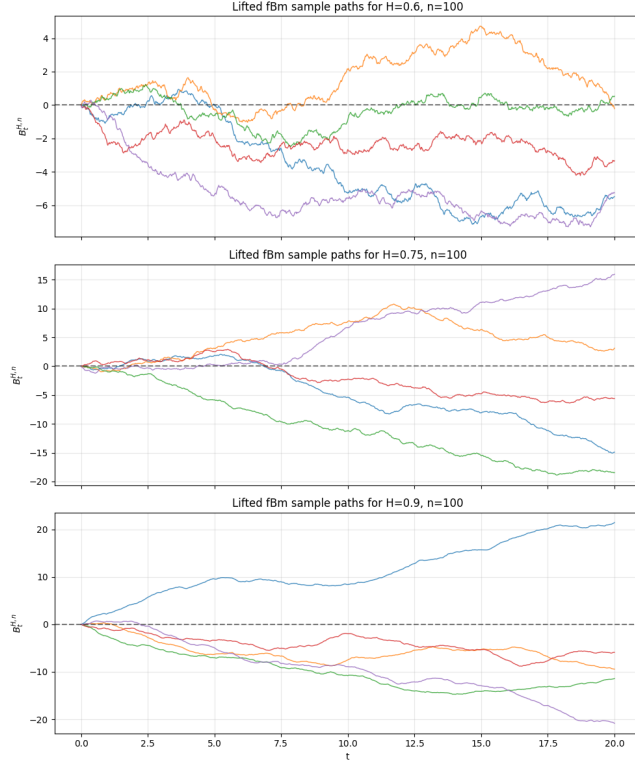


Figure 2.3: Sample paths of lifted fractional Brownian motion for $H = 0.6, 0.75, 0.9$ and $n = 100$ beta nodes. Implementation available at [5].

$\frac{1}{2}(t^{2H} + s^{2H} - |t - s|^{2H})$, across $H = 0.6, 0.75, 0.9$, using $n = 100$ nodes and $T = 20$. The implementation can be found at [5].

Figure 2.4 compares empirical and theoretical variance t^{2H} for all three values of H . The empirical curves track theory closely across the full range in every case.

Figure 2.5 compares the full theoretical and empirical covariance matrices, together with the pointwise relative error $|\text{Cov}_{\text{emp}} - \text{Cov}_{\text{true}}|/|\text{Cov}_{\text{true}}|$. The theoretical and empirical matrices are visually indistinguishable in all three cases. Mean relative covariance error was calculated in [5] to be 4.33% ($H = 0.6$), 2.55% ($H = 0.75$), and 1.03% ($H = 0.9$). The largest relative errors are concentrated at short lags, where the true covariance is smallest in magnitude and so most sensitive to small absolute discrepancies.

These plots are remarkably accurate, with the variance looking almost indistinguishable from theoretical variance, and our covariance heat map of the error being remarkably small as discussed. Given that fBm_H is defined as a Gaussian process with this covariance structure, this gives very strong evidence towards the $H > \frac{1}{2}$ lifted fBm model's convergence.

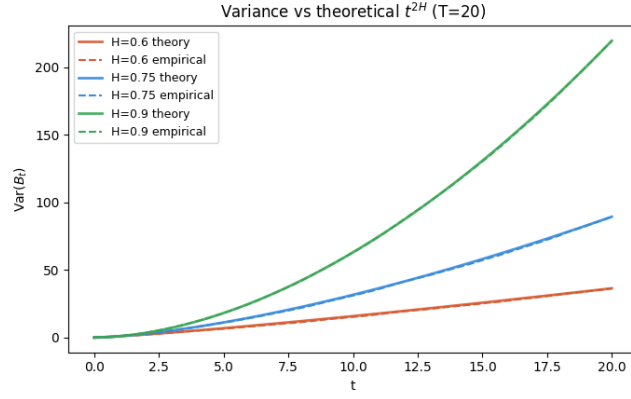


Figure 2.4: Empirical vs. theoretical variance t^{2H} for $H = 0.6, 0.75, 0.9$ ($T = 20$, $n = 100$ nodes).

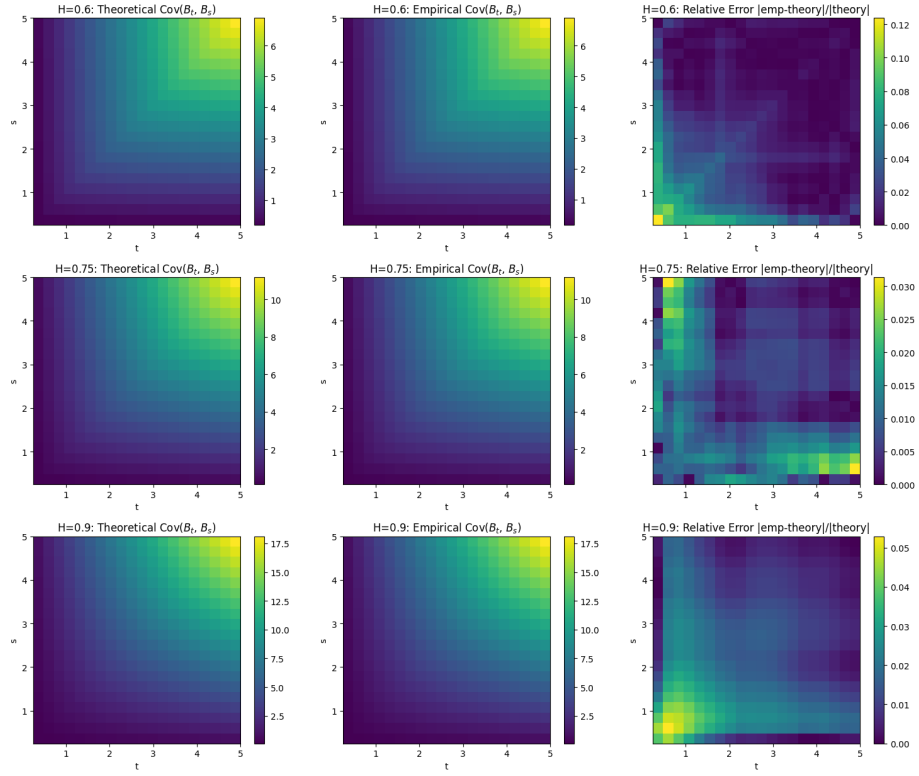


Figure 2.5: Theoretical covariance (left), empirical covariance (middle), and pointwise relative error (right), for $H = 0.6, 0.75, 0.9$ (top to bottom).

Chapter 3

Rough Volatility Modelling

Unlike Chapters 1-2, where we rigorously develop theorems and construct an approximation method in a novel setting, this chapter is deliberately expository: our aim is to situate that machinery within the literature that motivates it, not to derive new results nor go into the rigorous proofs of such results.

In this chapter we give a rapid introduction to volatility modelling, to give the reader a sense of how the exact same ideas and tools from our lifted fBm_H is used in the lifted Heston model. In the following chapter we assume that the reader has a basic working knowledge of financial mathematics and standard financial instruments.

Now that we have developed the theory of fractional Brownian motion, we are ready to apply it to a modern and active area of volatility modelling. We begin this chapter with the classical Black-Scholes model, which we show is insufficient and unrealistic as a description of how option prices behave in practice. This motivates stochastic volatility models, and in particular the Heston model, whose relatively simple characteristic function alongside realistic fits for the implied volatility surface have made it one of the most popular models in the literature.

However, in [8] the authors show that across a wide range of assets, the historical volatility process is significantly rougher than the smooth diffusive paths produced by a standard Brownian motion - with estimated Hurst exponents of order $H \approx 0.1$, far below the $H = 1/2$ (BM) assumptions. Motivated by this, practitioners have increasingly replaced the Heston model with a rough counterpart, which drives the variance process using a fractional kernel of exactly the form appearing in the Mandelbrot-Van Ness representation of fBm_H (1.3), rather than a standard Brownian increment.

This substitution comes at a cost since the resulting variance process inherits exactly the issues we proved for fBm_H itself in Chapter 1: it is neither a semimartingale (Theorem 1.2.3) nor a Markov process (Theorem 1.2.4). This is a serious practical obstacle, since standard tools for simulation and pricing - Markov-chain-based numerical schemes, Feynman-Kac arguments - rely fundamentally on the Markov property.

We resolve this using the Muravlev representation (1.4.4) developed in Chapter 1, together with its finite-dimensional approximation. This allows us to replace the fractional kernel with a finite superposition of Ornstein-Uhlenbeck kernels, yielding the so-called lifted or Markovian approximation of the rough Heston model - a model which is genuinely Markovian and hence tractable, while still capturing the empirically observed roughness of volatility.

3.1 Black-Scholes & Implied Volatility

A key assumption underlying most asset pricing is that log-price of an asset is typically modelled as a continuous semi-martingale¹, and thus² the asset price itself is the solution to an SDE of the form

$$\frac{dS_t}{S_t} = \mu_t dt + \sigma_t dB_t, \quad (3.1)$$

where μ_t is the drift, σ_t the volatility process, and B_t a standard Brownian motion. The choice of volatility process σ_t and the drift process μ_t is what distinguishes the models we will consider, and it is central to how well a model's prices match those observed in the market.

The simplest choice, taken by Black and Scholes [3], is to fix $\sigma_t = \sigma$ and $\mu_t = r$ constant, giving the classical geometric Brownian motion model

$$dS_t = rS_t dt + \sigma S_t dW_t, \quad S_0 > 0,$$

with r the (constant) risk-free rate³. This yields the closed-form Black-Scholes price for a European call option (see [2]),

$$C^{BS}(\sigma) = S_t \Phi(d_+) - K e^{-r(T-t)} \Phi(d_-), \quad d_{\pm} = \frac{\ln(S_t/K) + \left(r \pm \frac{\sigma^2}{2}\right)(T-t)}{\sigma \sqrt{T-t}}.$$

Where $\Phi(\cdot)$ denotes the standard normal cumulative distribution function; T is the time to maturity; K is the strike price price.

Note that C^{BS} is strictly increasing in σ (see [2]), any observed market price $C(K, T)$ for a European option with strike K and maturity T has a unique implied volatility $\sigma^{BS}(K, T)$ solving $C(K, T) = C^{BS}(K, T, \sigma^{BS}(K, T))$ - the volatility that, plugged into the Black-Scholes formula, reproduces the observed price exactly. Computing $\sigma^{BS}(K, T)$ across a grid of strikes and maturities produces the implied volatility surface, which is the standard target against which volatility models are calibrated: a model is considered consistent with the market if the implied volatility surface it generates matches the one observed empirically.

¹Recall from 1.2.3 that semi-martingales are the widest class of processes that are integrable in the Ito .

²We can derive equation 3.1 directly by noting that $X_t = \log(S_t)$ is a semi martingale and then applying Ito's formula $f(X_t) = e^{X_t}$.

³This risk free rate ensures that the process discounted to present time is a martingale.

If the Black-Scholes model's constant- σ assumption held in practice, this surface would be flat. It is not: real implied volatility surfaces exhibit both a smile (implied volatility rising for strikes far from the money) and a skew (asymmetry between low and high strikes), varying with both strike and maturity (see [9]). Since Black-Scholes cannot generate any of this variation, it systematically misprices options away from at-the-money, motivating the stochastic volatility models - beginning with the Heston model of Section 3.2 - that we develop for the remainder of this chapter.

3.2 The Heston Model

So clearly the Black-Scholes assumptions of constant volatility are highly unrealistic⁴. Thus people nowadays instead model the volatility as a stochastic process too. One of the most realistic of these stochastic volatility models is the Heston model, which is the key model we will be working with throughout this project, and has the following dynamics:

$$\begin{cases} dS_t = S_t \sqrt{v_t} dW_t \\ dv_t = \lambda(\theta - v_t) dt + \sigma \sqrt{v_t} dB_t \end{cases} \quad (3.2)$$

Here the initial value of the underlying asset S_t can be observed from the market while the parameters $\lambda, \theta, \sigma, v_t$ are positive and represent respectively the mean reversion rate, the long run variance, the volatility of the variance and the value of the variance process. Moreover we assume that the risk-free rate $r = 0$ and that the Brownian motions have a correlation factor $\rho = \langle W, B \rangle$.

This model also makes sense from a heuristic standpoint. First of all the asset process agrees with the empirical observation that over small increments the stock price changes proportionally to the square root of the variance. Moreover under this model we have that the assumption that the volatility of variance σ is constant, notice that the variance process is looks very similar to the OU process from section 1.4.1 with the additional $\sqrt{v_t}$ term which makes it an example of a CIR process⁵. The Heston model is very popular due to the fact that it generates reasonable shapes and dynamics for the implied volatility surface and because there is an explicit formula for the characteristic function of the asset logprice. For this reason it is possible to calibrate the model to European options in a very efficient way and then to use the calibrated model in order to price exotic options such as American or Asian-type performing for example a Monte Carlo simulation.

Theorem 3.2.1 (Characteristic Function of Classical Heston Model). *Consider the Heston model in Equation 3.2, and define $X_t = \log(S_t)$ and consider the natural filtration $\mathcal{F}_t = \sigma(W_s, B_s \mid 0 \leq s \leq t \leq T)$ it holds true that*

$$F(t, z, X_t, v_t) = \mathbb{E}(e^{izX_T} \mid \mathcal{F}_t) = e^{A(t) + B(t)v_t + C(t)X_t}$$

⁴This would imply that volatility surfaces would be flat, which is not the case empirically.

⁵One additional constraint is that the feller condition: $2\lambda\theta > \sigma^2$ must be satisfied. This ensures that the variance remains positive.

Proof. First of all we apply Itô's Lemma to $X_t = \log(S_t)$, then we get

$$dX_t = -\frac{v_t}{2}dt + \sqrt{v_t}dW_t$$

If we apply the Feynmann-Kac formula for the function $F(t, z, X_t, v_t) = \mathbb{E}(e^{izX_T} | \mathcal{F}_t)$ we find the PDE:

$$\begin{cases} \frac{\partial F}{\partial t} - \frac{v}{2} \frac{\partial F}{\partial x} + \lambda(\theta - v) \frac{\partial F}{\partial v} + \frac{v}{2} \frac{\partial^2 F}{\partial x^2} + \frac{\sigma^2 v}{2} \frac{\partial^2 F}{\partial v^2} + \sigma \rho v \frac{\partial^2 F}{\partial x \partial v} = 0 \\ F(x, v, T) = e^{izx} \end{cases}$$

Supposing that $F(t, z, X_t, v_t) = e^{A(t)+B(t)v_t+C(t)X_t}$ and imposing that it is a solution we have:

$$\begin{cases} F \left[(\dot{A} + \dot{B}v + \dot{C}x) - \frac{v}{2}C + \lambda(\theta - v)B + \frac{v}{2}C^2 + \frac{\sigma^2 v}{2}B^2 + \sigma \rho v BC \right] = 0 \\ A(T) = B(T) = 0 \\ C(T) = iz \end{cases}$$

We can simplify F so this is a Riccati differential equation that can be solved analytically and so the theorem is proved $\square$

Moreover, as we said above, in [8], the authors show that for a wide range of assets the historical volatility is much rougher than the one reproduced by a standard Brownian motion. Due to this it has become common to replace the standard Heston model with its rough counterpart using a fractional kernel identical to the one used in the Mandelbrot-Van Ness representation.

3.3 Volatility is Rough

The classical assumption - implicit in both Black-Scholes and the standard Heston model - is that volatility evolves as a diffusion driven by ordinary Brownian motion, and hence has Hölder regularity $\frac{1}{2}$. [8] tested this assumption directly against historical data: examining log-volatility time series across a wide range of assets, they estimated the Hölder exponent of the paths. Across every asset studied, the estimated exponent was far below $\frac{1}{2}$ - typically $H \approx 0.1$ - indicating that volatility paths are considerably rougher than Brownian motion, not smoother. This finding, robust across asset classes and time periods, motivates replacing the Brownian-driven variance process of the Heston model with one driven by a fractional kernel of Hurst parameter $H \approx 0.1$, giving the rough Heston model which we will explore next.

3.4 The Rough Heston Model

The rough Heston model (RHM) is built up in such a way that the underlying asset S follows the same dynamics of the classical Heston model with the variance v defined to be H-Hölder continuous:

$$\left\{ \begin{aligned} dS_t &= S_t \sqrt{v_t} dW_t, \\ v_t &= v_0 + \frac{1}{\Gamma(H + \frac{1}{2})} \int_0^t (t-s)^{H-\frac{1}{2}} \lambda(\theta - v_s) ds \\ &\quad + \frac{1}{\Gamma(H + \frac{1}{2})} \int_0^t (t-s)^{H-\frac{1}{2}} \sigma \sqrt{v_s} dB_s \\ &= v_0 + I_0^{H+\frac{1}{2}} [\lambda(\theta - v_s) ds + \sigma \sqrt{v_s} dB_s], \end{aligned} \right. \quad (3.3)$$

where $H \in (0, \frac{1}{2})$. Notice that the idea here is that we are fractionally integrating the Heston's volatility process in such a way that we have paths that are H-Hölder continuous thereby giving us the rough paths that agree with empirical observations. The fact that the volatility in the rough Heston model is no more a semimartingale and a Markov process seems to be a problem for its tractability. However just like in the standard Heston case, it was demonstrated in [6] that we can (semi) explicit write down the characteristic function for the rough Heston model, where our solution requires us to solve a related fractional Riccati equation.

Theorem 3.4.1 (Characteristic Function of Rough Heston Model). *Consider the rough Heston model given by 3.3 and define $X_t = \log(S_t)$ then we have*

$$\mathbb{E}(e^{iaX_t}) = \exp\left(\theta \lambda I^1 h(a, t) + v_0 I^{\frac{1}{2}-H} h(a, t)\right)$$

where $h(a, \cdot)$ is the solution of the fractional Riccati equation

$$\begin{cases} D^{H+\frac{1}{2}} h(a, t) = \frac{1}{2}(-a^2 - ia) + (ia\rho\sigma - \lambda)h(a, t) + \frac{\sigma^2}{2}h^2(a, t) \\ I^{\frac{1}{2}-H} h(a, 0) = 0 \end{cases}$$

We choose not to include the long derivation of this result which can be found in [6]. The main idea behind the proof is to show that the model can be written as a limiting sequence of a process known as a Hawks process. We then find the characteristic function of each of the Hawks processes and show that this converges to our characteristic function as required. As we discussed in the standard Heston case, the characteristic function is only useful for finding the value of the model at the final time step which is useful for the calibration of the model and for pricing European options, but it is not possible to use it for pricing exotic options which require Monte-Carlo simulation of the full asset's path. Nevertheless due to the lack of Markovianity of the volatility process the simulation is difficult, thus in order to simulate the Rough heston paths we instead introduce the lifted Heston model from [10].

3.4.1 Markovian Approximation of Rough Heston

In this section we only consider the case $H \in (0, \frac{1}{2})$, consistent with the empirically observed value $H \approx 0.1$ discussed in Section 3.3. To resolve the non-

Markovianity issue identified above, we follow exactly the strategy of the Muravlev representation developed in Chapter 1: rather than work with the non-Markovian, singular kernel directly, we approximate it by a finite superposition of exponential OU-type kernels, each of which is Markovian, and for which the aggregate system remains Markovian as a whole. Following [10], for a fixed $n \in \mathbb{N}$ we define the lifted Heston model by

$$dS_t^n = S_t^n \sqrt{v_t^n} dW_t, \quad S_0^n \geq 0, \quad (3.4)$$

where the lifted variance process v_t^n is built from n underlying factor processes $U^{n,1}, \dots, U^{n,n}$ via

$$v_t^n = g_0^n(t) + \sum_{i=1}^n w_i^n U_t^{n,i}, \quad (3.5)$$

$$dU_t^{n,i} = \left(-\beta_i^n U_t^{n,i} - \lambda v_t^n \right) dt + \sigma \sqrt{v_t^n} dB_t, \quad U_0^{n,i} = 0, \quad i = 1, \dots, n, \quad (3.6)$$

with deterministic function g_0^n , constants $\lambda, \sigma \in \mathbb{R}^+$, weights and mean-reversion speeds $w_i^n, \beta_i^n \geq 0$, and correlation $\rho dt = d\langle B, W \rangle_t$, $\rho \in [-1, 1]$.

At first glance, (3.15) does not resemble the Ornstein–Uhlenbeck process of Section 1.4.1: there, each factor Z_t^β was driven by an independent Brownian increment with constant diffusion coefficient, and the factors did not interact. Here, two structural differences appear. First, the diffusion coefficient $\sigma \sqrt{v_t^n}$ is state-dependent rather than constant – the same CIR-type square-root diffusion that distinguishes the classical Heston variance process from a standard OU process (Section 3.2). Second, the drift of each factor $U_t^{n,i}$ contains an additional term $-\lambda v_t^n$, beyond the usual mean-reversion $-\beta_i^n U_t^{n,i}$. Crucially, v_t^n in (3.14) is the weighted sum of all n factors, so every factor’s dynamics in (3.15) depend on the entire system, not on $U_t^{n,i}$ in isolation. The factors $U^{n,1}, \dots, U^{n,n}$ are therefore not independent OU processes, but a correlated system coupled through the shared variance v_t^n . The extra drift term $-\lambda v_t^n$ is precisely what is needed so that, once the factors are recombined via the weights w_i^n in (3.14), the combined process v_t^n reproduces the correct mean-reverting behaviour of the rough Heston variance process as a whole.

In [10], existence and uniqueness of (3.13)–(3.15) is established rigorously. This approximation of the RHM preserves the tractability of the classical Heston model while retaining the roughness of its rough counterpart, and substantially speeds up simulation.

Taking expectations on both sides of (3.14) and observing⁶ that $\mathbb{E}(U_t^{n,i}) = \int_0^t -\lambda e^{-\beta_i^n(t-s)} \mathbb{E}(v_s^n) ds$ yields the relation

$$g_0^n(t) = \mathbb{E}(v_t^n) + \lambda \sum_{i=1}^n w_i^n \int_0^t e^{-\beta_i^n(t-s)} \mathbb{E}(v_s^n) ds. \quad (3.7)$$

⁶This follows from Itô’s lemma applied to $f(t, x) = e^{\beta_i^n t} x$, using the integrating factor $e^{\beta_i^n t}$ to eliminate the mean-reversion term in (3.15); taking expectations then removes the stochastic integral, which is a martingale.

We restrict attention⁷ to the case

$$g_0^n(t) = v_0 + \lambda\theta \sum_{i=1}^n w_i^n \int_0^t e^{-\beta_i^n(t-s)} ds, \quad v_0, \theta \geq 0. \quad (3.8)$$

Notice that $n = 1$, $w_1^1 = 1$, $\beta_1^1 = 0$ recovers the classical Heston model exactly. The lifted model also admits a closed-form characteristic function:

Theorem 3.4.2 (Characteristic function for lifted Heston). *Consider the lifted Heston model (3.13)–(3.15) and define $X_t^n = \log(S_t^n)$. Then*

$$\mathbb{E}(e^{iaX_T^n} | \mathcal{F}_t) = \exp \left(\phi^n(t, T) + iaX_t^n + \sum_{i=1}^n w_i^n \psi^{n,i}(T-t) U_t^{n,i} \right), \quad \forall t \leq T,$$

where $(\psi^{n,i})_{1 \leq i \leq n}$ solves the n -dimensional system of Riccati ODEs

$$(\psi^{n,i})' = -\beta_i^n \psi^{n,i} + F \left(ia, \sum_{j=1}^n w_j^n \psi^{n,j} \right), \quad \psi^{n,i}(0) = 0, \quad i = 1, \dots, n,$$

with

$$F(ia, v) = \frac{1}{2}(-a^2 - ia) + (\rho\sigma ia - \lambda)v + \frac{\sigma^2}{2}v^2, \quad \phi^n(t, T) = \int_0^{T-t} F \left(ia, \sum_{i=1}^n w_i^n \psi^{n,i}(s) \right) g_0^n(T-s) ds.$$

A proof sketch is given in Appendix ??.

This system carries $2n + 5$ parameters: five from $(v_0, \lambda, \theta, \rho, \sigma)$ and $2n$ from $(w_i^n, \beta_i^n)_{i=1, \dots, n}$. In [10], however, $(w_i^n, \beta_i^n)_{i=1, \dots, n}$ is parameterized in terms of the target Hurst index H and a single additional parameter r_n – exactly the scheme of equations (2.6), which we adapted directly to fBm itself in Chapter 1. Once H is fixed (in practice $H \approx 0.1$, per Section 3.3), this reduces the parameter count to six.

We recall the fractional kernel identity of Theorem 1.4.1 in an equivalent form, to make the convergence argument below easier to follow:

Theorem 3.4.3 (Fractional kernel for $H \in (0, \frac{1}{2})$). *The fractional kernel appearing in (3.3) admits the representation*

$$K(t) = \frac{t^{H-1/2}}{\Gamma(H + \frac{1}{2})} = \int_0^\infty e^{-\beta t} \mu(d\beta), \quad \mu(d\beta) = \frac{\beta^{-H-1/2} d\beta}{\Gamma(H + \frac{1}{2})\Gamma(\frac{1}{2} - H)}. \quad (3.9)$$

This is the same identity proved in Theorem 1.4.1, restated here for convenience. Setting $v_0 = 0$, this lets us rewrite the rough Heston variance (3.3) as a

⁷This is precisely the case ensuring $n = 1$ recovers the classical Heston model, so the lifted model is a genuine generalization that converges to the rough model rather than an unrelated approximation.

superposition of infinitely many processes with identical dynamics but distinct mean-reversion rates:

$$v_t = \int_0^\infty U_t^\beta \mu(d\beta), \quad (3.10)$$

$$U_t^\beta = \int_0^t e^{-\beta(t-s)} \left(\lambda(\theta - v_s) ds + \sigma \sqrt{v_s} dB_s \right), \quad \beta > 0. \quad (3.11)$$

3.4.2 Convergence to Rough Heston Model

The key result of this section is that the lifted Heston model converges to rough Heston as $n \rightarrow \infty$ in the L^2 sense. Using (3.15), equation (3.14) may be rewritten as a stochastic Volterra equation,

$$v_t^n = g_0^n(t) + \int_0^t K^n(t-s) \left(-\lambda v_s^n ds + \sigma \sqrt{v_s^n} dB_s \right), \quad K^n(t) := \sum_{i=1}^n w_i^n e^{-\beta_i^n t}. \quad (3.12)$$

This is the key link between the theory developed in Chapters 1–2 and financial modelling: with (w_i^n, β_i^n) chosen as in Theorem 2.2.1, $K^n \rightarrow K$ in $L^2(0, T)$ is exactly the kernel convergence result proved there. Kernel convergence alone is not, however, sufficient to conclude $v_t^n \rightarrow v_t$: since v_t^n solves a Volterra equation with K^n appearing as an input rather than as an explicit function of v_t^n , we additionally need the map from kernel to solution to be *stable* – that small perturbations of the kernel produce correspondingly small perturbations of the solution. This stability is established in [10] via a Grönwall-type argument, which we take as given rather than reproduce. Combining kernel convergence with this stability result yields $v_t^n \rightarrow v_t$ uniformly on $[0, T]$ in probability, and hence, since S_t^n is built directly from v_t^n , that $S_t^n \rightarrow S_t$: the lifted Heston price process converges to the true rough Heston price process as $n \rightarrow \infty$.

3.5 The Lifted Heston Model

3.5.1 Markovian Approximation of Rough Heston

In this section we only consider the case $H \in (0, \frac{1}{2})$, consistent with the empirically observed value $H \approx 0.1$ discussed in Section 3.3. To resolve the non-Markovianity issue identified above, we follow exactly the strategy of the Muravlev representation developed in Chapter 1: rather than work with the non-Markovian, singular kernel directly, we approximate it by a finite superposition of exponential OU-type kernels, each of which is Markovian, and for which the aggregate system remains Markovian as a whole. Following [10], for a fixed $n \in \mathbb{N}$ we define the lifted Heston model by

$$dS_t^n = S_t^n \sqrt{v_t^n} dW_t, \quad S_0^n \geq 0, \quad (3.13)$$

where the lifted variance process v_t^n is built from n underlying factor processes $U^{n,1}, \dots, U^{n,n}$ via

$$v_t^n = g_0^n(t) + \sum_{i=1}^n w_i^n U_t^{n,i}, \quad (3.14)$$

$$dU_t^{n,i} = \left(-\beta_i^n U_t^{n,i} - \lambda v_t^n \right) dt + \sigma \sqrt{v_t^n} dB_t, \quad U_0^{n,i} = 0, \quad i = 1, \dots, n, \quad (3.15)$$

with deterministic function g_0^n , constants $\lambda, \sigma \in \mathbb{R}^+$, weights and mean-reversion speeds $w_i^n, \beta_i^n \geq 0$, and correlation $\rho dt = d\langle B, W \rangle_t$, $\rho \in [-1, 1]$.

At first glance, (3.15) does not resemble the Ornstein–Uhlenbeck process of Section 1.4.1: there, each factor Z_t^β was driven by an independent Brownian increment with constant diffusion coefficient, and the factors did not interact. Here, two structural differences appear. First, the diffusion coefficient $\sigma \sqrt{v_t^n}$ is state-dependent rather than constant – the same CIR-type square-root diffusion that distinguishes the classical Heston variance process from a standard OU process (Section 3.2). Second, the drift of each factor $U_t^{n,i}$ contains an additional term $-\lambda v_t^n$, beyond the usual mean-reversion $-\beta_i^n U_t^{n,i}$. Crucially, v_t^n in (3.14) is the weighted sum of all n factors, so every factor’s dynamics in (3.15) depend on the entire system, not on $U_t^{n,i}$ in isolation. The factors $U^{n,1}, \dots, U^{n,n}$ are therefore not independent OU processes, but a correlated system coupled through the shared variance v_t^n . The extra drift term $-\lambda v_t^n$ is precisely what is needed so that, once the factors are recombined via the weights w_i^n in (3.14), the combined process v_t^n reproduces the correct mean-reverting behaviour of the rough Heston variance process as a whole.

In [10], existence and uniqueness of (3.13)–(3.15) is established rigorously. This approximation of the RHM preserves the tractability of the classical Heston model while retaining the roughness of its rough counterpart, and substantially speeds up simulation.

Taking expectations on both sides of (3.14) and observing⁸ that $\mathbb{E}(U_t^{n,i}) = \int_0^t -\lambda e^{-\beta_i^n(t-s)} \mathbb{E}(v_s^n) ds$ yields the relation

$$g_0^n(t) = \mathbb{E}(v_t^n) + \lambda \sum_{i=1}^n w_i^n \int_0^t e^{-\beta_i^n(t-s)} \mathbb{E}(v_s^n) ds. \quad (3.16)$$

We restrict attention⁹ to the case

$$g_0^n(t) = v_0 + \lambda \theta \sum_{i=1}^n w_i^n \int_0^t e^{-\beta_i^n(t-s)} ds, \quad v_0, \theta \geq 0. \quad (3.17)$$

Notice that $n = 1$, $w_1^1 = 1$, $\beta_1^1 = 0$ recovers the classical Heston model exactly. The lifted model also admits a closed-form characteristic function:

⁸This follows from Itô’s lemma applied to $f(t, x) = e^{\beta_i^n t} x$, using the integrating factor $e^{\beta_i^n t}$ to eliminate the mean-reversion term in (3.15); taking expectations then removes the stochastic integral, which is a martingale.

⁹This is precisely the case ensuring $n = 1$ recovers the classical Heston model, so the lifted model is a genuine generalization that converges to the rough model rather than an unrelated approximation.

Theorem 3.5.1 (Characteristic function for lifted Heston). *Consider the lifted Heston model (3.13)–(3.15) and define $X_t^n = \log(S_t^n)$. Then*

$$\mathbb{E}(e^{iaX_t^n} | \mathcal{F}_t) = \exp \left(\phi^n(t, T) + iaX_t^n + \sum_{i=1}^n w_i^n \psi^{n,i}(T-t) U_t^{n,i} \right), \quad \forall t \leq T,$$

where $(\psi^{n,i})_{1 \leq i \leq n}$ solves the n -dimensional system of Riccati ODEs

$$(\psi^{n,i})' = -\beta_i^n \psi^{n,i} + F \left(ia, \sum_{j=1}^n w_j^n \psi^{n,j} \right), \quad \psi^{n,i}(0) = 0, \quad i = 1, \dots, n,$$

with

$$F(ia, v) = \frac{1}{2}(-a^2 - ia) + (\rho\sigma ia - \lambda)v + \frac{\sigma^2}{2}v^2, \quad \phi^n(t, T) = \int_0^{T-t} F \left(ia, \sum_{i=1}^n w_i^n \psi^{n,i}(s) \right) g_0^n(T-s) ds.$$

A proof sketch is given in Appendix ??.

This system carries $2n + 5$ parameters: five from $(v_0, \lambda, \theta, \rho, \sigma)$ and $2n$ from $(w_i^n, \beta_i^n)_{i=1, \dots, n}$. In [10], however, $(w_i^n, \beta_i^n)_{i=1, \dots, n}$ is parameterized in terms of the target Hurst index H and a single additional parameter r_n - exactly the scheme of equations (2.6), which we adapted directly to fBm itself in Chapter 1. Once H is fixed (in practice $H \approx 0.1$, per Section 3.3), this reduces the parameter count to six.

We recall the fractional kernel identity of Theorem 1.4.1 in an equivalent form, to make the convergence argument below easier to follow:

Theorem 3.5.2 (Fractional kernel for $H \in (0, \frac{1}{2})$). *The fractional kernel appearing in (3.3) admits the representation*

$$K(t) = \frac{t^{H-1/2}}{\Gamma(H + \frac{1}{2})} = \int_0^\infty e^{-\beta t} \mu(d\beta), \quad \mu(d\beta) = \frac{\beta^{-H-1/2} d\beta}{\Gamma(H + \frac{1}{2})\Gamma(\frac{1}{2} - H)}. \quad (3.18)$$

This is the same identity proved in Theorem 1.4.1, restated here for convenience. Setting $v_0 = 0$, this lets us rewrite the rough Heston variance (3.3) as a superposition of infinitely many processes with identical dynamics but distinct mean-reversion rates:

$$v_t = \int_0^\infty U_t^\beta \mu(d\beta), \quad (3.19)$$

$$U_t^\beta = \int_0^t e^{-\beta(t-s)} \left(\lambda(\theta - v_s) ds + \sigma \sqrt{v_s} dB_s \right), \quad \beta > 0. \quad (3.20)$$

3.5.2 Convergence to Rough Heston Model

The key result of this section is that the lifted Heston model converges to rough Heston as $n \rightarrow \infty$ in the L^2 sense. Using (3.15), equation (3.14) may be rewrit-

ten as a stochastic Volterra equation,

$$v_t^n = g_0^n(t) + \int_0^t K^n(t-s) (-\lambda v_s^n ds + \sigma \sqrt{v_s^n} dB_s),$$

$$K^n(t) := \sum_{i=1}^n w_i^n e^{-\beta_i^n t}.$$

This is the key link between the theory developed in Chapters 1-2 and financial modelling: with (w_i^n, β_i^n) chosen as in Theorem 2.2.1, $K^n \rightarrow K$ in $L^2(0, T)$ is exactly the kernel convergence result proved there. Kernel convergence alone is not, however, sufficient to conclude $v_t^n \rightarrow v_t$: since v_t^n solves a Volterra equation with K^n appearing as an input rather than as an explicit function of v_t^n , we additionally need the map from kernel to solution to be stable - that small perturbations of the kernel produce correspondingly small perturbations of the solution. This stability is established in [10] via a Grönwall-type argument, which we take as given rather than reproduce. Combining kernel convergence with this stability result yields $v_t^n \rightarrow v_t$ uniformly on $[0, T]$ in probability, and hence, since S_t^n is built directly from v_t^n , that $S_t^n \rightarrow S_t$: the lifted Heston price process converges to the true rough Heston price process as $n \rightarrow \infty$.

Chapter 4

Conclusion & Future Developments

This project set out to give a self-contained, constructive treatment of fractional Brownian motion suitable for an advanced undergraduate reader, culminating in its application to rough volatility modelling. Starting from fractional calculus and the covariance structure of fBm_H , we developed the Mandelbrot-Van Ness and Volterra kernel representations before arriving at the infinite-dimensional Ornstein-Uhlenbeck (Muravlev) representation, which resolves the central practical obstacle in working with fBm_H : its lack of the Markov property.

Adapting the node and weight parametrisation underlying the lifted Heston model of [10] directly to the fBm kernel, we constructed a finite-dimensional Markovian approximation of fBm_H itself - *lifted fBm_H* - after first identifying, and empirically demonstrating, a genuine failure mode of the naive fixed-ratio geometric grid we initially attempted. For the sub-diffusive regime $H < \frac{1}{2}$, we went beyond construction and validation: we proved that lifted fBm_H converges to true, stationary-increment fBm in $L^2(\Omega)$. This required first establishing convergence of the discretised fractional kernel itself, then lifting this to the full stochastic process via the Itô isometry - a reduction that, while structurally clean, concealed a genuine subtlety in its compensator term, which we diagnosed and resolved in full in Appendix C. We then extended the construction to the super-diffusive regime $H > \frac{1}{2}$ via a regularised representation that subtracts the driving Brownian motion itself, using the same node scheme throughout. As far as we are aware no analogous convergence proof is currently available for this regime, and we identified precisely why: the underlying measure ceases to be locally finite near the origin once $H > \frac{1}{2}$, so the estimates underpinning our $H < \frac{1}{2}$ proof fail structurally rather than merely requiring sharper constants. In its place we gave strong empirical evidence - close agreement in both variance scaling and full covariance structure across a range of H values - that convergence nonetheless holds. Chapter 3 then situated this machinery within its motivating application, connecting the Markovian lifting technique to the

lifted Heston model and the broader rough volatility literature.

Several natural extensions were beyond the scope of what could be completed in the time available for this project.

On the simulation side, we would have liked to implement and compare the Davies-Harte circulant embedding method alongside the Cholesky method of Section 1.5.1, giving a fuller picture of the trade-offs between exact simulation schemes for fBm_H . Relatedly, it would have been valuable to apply Hurst parameter estimation techniques directly to historical financial data, to demonstrate first-hand - rather than by citing [8] - that realised volatility exhibits the roughness properties motivating the rough Heston model.

On the theoretical side, the most pressing open question raised by this report is a rigorous convergence proof for lifted fBm_H in the $H > \frac{1}{2}$ regime, analogous to Theorem 2.2.2 for $H < \frac{1}{2}$. As discussed in Section 2.3, this is not simply a matter of adapting existing estimates: a correct proof would need to exploit the additional cancellation introduced by the $-B_t$ correction term to compensate for the loss of local finiteness of the underlying measure near the origin, most likely yielding convergence rates genuinely different from - and probably slower than - those obtained for $H < \frac{1}{2}$. More broadly, several theorems throughout Chapter 1 were presented with sketched or omitted proofs where the full argument, while important, was judged secondary to the main constructive thread of the work. In this sense, the classical background material of this report remains closer to a guided survey than an exhaustive proof-by-proof treatment, even though its original contribution - lifted fBm_H for $H < \frac{1}{2}$ - is now supported by a complete convergence proof; a more complete treatment would fill in the remaining sketched results and close the $H > \frac{1}{2}$ gap identified above.

Finally, we deliberately restricted our attention to fBm_H itself and its Markovian lifting, rather than the stochastic Volterra equations underlying rough Heston more directly. A fuller treatment would develop the theory of stochastic integration against Volterra kernels in its own right, which would in turn better motivate the specific structure of the rough Heston variance process. It would also be natural to extend the discussion to other rough volatility models beyond rough Heston, such as the rough Bergomi model, to give a broader view of how the Markovian lifting technique developed here applies across the rough volatility literature.

Appendix A

Simulating the OU Process

A full implementation of the OU process using the following ideas can be found at [5]. Recall the standard OU process with mean reversion β ,

$$dZ_t^\beta = -\beta Z_t^\beta dt + dB_t,$$

introduced in Section 1.4.1. Since the OU process is Gaussian and Markovian, its transition law is known exactly: for $s < t$,

$$Z_t^\beta \mid Z_s^\beta \sim \mathcal{N}\left(e^{-\beta(t-s)} Z_s^\beta, \frac{1 - e^{-2\beta(t-s)}}{2\beta}\right).$$

This gives an exact (as opposed to Euler-discretized) simulation scheme on any time grid: for a step Δt ,

$$Z_{t+\Delta t}^\beta = e^{-\beta\Delta t} Z_t^\beta + \sqrt{\frac{1 - e^{-2\beta\Delta t}}{2\beta}} W, \quad W \sim \mathcal{N}(0, 1),$$

which introduces no discretization error regardless of the step size Δt , unlike a standard Euler scheme.

For the lifted fBm construction, we require the process to be stationary - recall $Z_t^\beta = \int_{-\infty}^t e^{-\beta(t-u)} dB_u$ - rather than started from $Z_0^\beta = 0$. The stationary marginal is $Z_0^\beta \sim \mathcal{N}\left(0, \frac{1}{2\beta}\right)$, and moreover, since each $Z_t^{\beta_j}$ is driven by the same underlying Brownian motion B , the initial values across different mean-reversion rates are correlated:

$$\text{Cov}\left(Z_0^{\beta_i}, Z_0^{\beta_j}\right) = \int_{-\infty}^0 e^{\beta_i u} e^{\beta_j u} du = \frac{1}{\beta_i + \beta_j}.$$

Writing $\Gamma_{i,j} = \frac{1}{\beta_i + \beta_j}$ for this covariance matrix across the quadrature nodes $\beta_{-n+1}, \dots, \beta_n$, we simulate the correlated stationary initial vector via a Cholesky decomposition $\Gamma = LL^\top$: drawing $\xi \sim \mathcal{N}(0, I)$ and setting $Z_0 = L\xi$ gives Z_0

the correct joint distribution. Each subsequent time step is then simulated independently across j using the exact recursion above, sharing the same driving Brownian increment W across all β_j at each step (since all $Z_t^{\beta_j}$ are driven by the same B_t).

Appendix B

Lifted Heston Characteristic Function

Proof sketch. The argument follows the same Feynman-Kac strategy used in the proof of Theorem 3.2.1 (the classical Heston case), but applied to the $(n+1)$ -dimensional Markov process $(X_t^n, U_t^{n,1}, \dots, U_t^{n,n})$ rather than the two-dimensional pair (X_t, v_t) .

Applying Itô's lemma to $X_t^n = \log(S_t^n)$ using (3.13) and (3.15), and then the Feynman-Kac formula to

$$F(t, a, X_t^n, U_t^{n,1}, \dots, U_t^{n,n}) := \mathbb{E} \left[e^{iaX_T^n} \mid \mathcal{F}_t \right],$$

yields a linear PDE in $t, x, u_1, \dots, u_n$ analogous to the scalar case, but now with n first-order terms $\partial F / \partial u_i$ in place of the single term $\partial F / \partial v$. The second-order structure is correspondingly richer: a cross-term $\partial^2 F / \partial x \partial u_i$ appears for each factor - exactly as the single cross-term $\partial^2 F / \partial x \partial v$ arose in the classical case from $\rho = \langle W, B \rangle$, since every $U^{n,i}$ is driven by the same Brownian motion B that drives X_t^n - together with cross-terms $\partial^2 F / \partial u_i \partial u_j$ between every pair of factors, reflecting the coupling of all n factors through the shared variance $v_t^n = g_0^n(t) + \sum_{i=1}^n w_i^n u_i$ appearing in the drift and diffusion coefficients of every factor (see (3.15)).

Proposing the exponential-affine ansatz

$$F = \exp \left(\phi^n(t, T) + iaX_t^n + \sum_{i=1}^n w_i^n \psi^{n,i}(T-t) U_t^{n,i} \right),$$

motivated by the affine structure of (3.14)-(3.15) and by analogy with the ansatz $F = e^{A(t)+B(t)v_t+C(t)X_t}$ used in the scalar Heston case - assigning one coefficient function $\psi^{n,i}(t)$ to each factor in place of the single $B(t)$ - and substituting into the PDE, the coefficient of each $U^{n,i}$ produces one ODE in the system stated above, while matching the remaining (state-independent) terms produces the

integral expression for ϕ^n . The coupling between factors enters only through the common term $F\left(ia, \sum_j w_j^n \psi^{n,j}\right)$ appearing in every equation of the system, which is exactly the source of the nonlinear coupling identified in Section ?? . Terminal conditions $\psi^{n,i}(0) = 0$ and $\phi^n(T, T) = 0$ follow from $F(T, \cdot) = e^{iax}$ as in the scalar case.

We omit the full verification that this ansatz solves the PDE, as it is lengthy but mechanical; the complete derivation, together with existence and uniqueness for the resulting Riccati system, is given in [10] and follows the same programme used to derive the characteristic function of the rough Heston model in [6]. $\square$

Appendix C

Completing the L^2 Estimate for Lifted fBm Convergence

In Section 2.2 we reduced the convergence of lifted fBm to true fBm to bounding two deterministic quantities arising from the Itô isometry: the on-diagonal term $A_n(t)$, controlled directly by kernel convergence (Theorem 2.2.1), and another remaining term - arising from the $(-\infty, 0)$ piece of the Mandelbrot-Van Ness representation - which we denote here by $R_n(t)$ (to avoid confusion with $B_t^{H,n}$, the approximating process itself, and B_t , the driving Brownian motion). We recall the setup before proceeding.

Writing $\Delta^n := K^n - K$ for the difference between the discretised and true fractional kernels, the Itô isometry applied to our integral representation of $B_t^{H,n} - B_t^H$ gives

$$\mathbb{E}[(B_t^{H,n} - B_t^H)^2] = A_n(t) + R_n(t), \quad A_n(t) := \int_0^t \Delta^n(v)^2 dv,$$

$$R_n(t) := \int_{-\infty}^0 (\Delta^n(t-u) - \Delta^n(-u))^2 du = \int_0^\infty (\Delta^n(t+w) - \Delta^n(w))^2 dw, \quad (\text{C.1})$$

the second equality following from the substitution $w = -u$. While $A_n(t) \rightarrow 0$ is immediate from Theorem 2.2.1, $R_n(t)$ resists the same approach directly since a naive attempt to bound (C.1) by splitting $(\Delta^n(t+w) - \Delta^n(w))^2 \leq 2\Delta^n(t+w)^2 + 2\Delta^n(w)^2$ and invoking $L^2(0, \infty)$ convergence of $K^n \rightarrow K$ fails, since $K(t) = t^{H-1/2}/\Gamma(H + \frac{1}{2})$ decays too slowly to be square-integrable on $(0, \infty)$ at all. The resolution is to work with the difference $\Delta^n(t+w) - \Delta^n(w)$ directly, via its derivative - which, unlike Δ^n itself, decays fast enough to remain in L^2 once we stay away from the origin.

C.1 A derivative kernel and its convergence

The key realisation is that differentiating the kernel gains an extra power of β in the underlying measure, which improves the decay just enough. We define the derivative measure $\nu(d\beta) := \beta \mu(d\beta)$, and correspondingly

$$K'(t) = - \int_0^\infty \beta e^{-\beta t} \mu(d\beta) = - \int_0^\infty e^{-\beta t} \nu(d\beta).$$

Unlike K , the derivative $K'(t) \propto t^{H-3/2}$ is square-integrable away from the origin since $\int_1^\infty t^{2H-3} dt$ converges for every $H < 1$, since $2H - 3 < -1$. This is precisely the room we need. Before bounding K' against its discretised counterpart, we first check that the same nodes (β_i^n) used to discretise K also give the natural discretisation of K' (that no new node construction is needed).

Lemma C.1.1 (Weight identity for the derivative measure). Let $\theta_i^n := r_n^{i-n/2}$ denote the same grid points used in the construction of (w_i^n, β_i^n) in Section 2.1.¹ Then

$$w_i^n \beta_i^n = \nu([\theta_{i-1}^n, \theta_i^n]), \quad i = 1, \dots, n.$$

Proof. Since $\mu(d\beta) = \beta^{-\alpha} d\beta / (\Gamma(\alpha)\Gamma(1-\alpha))$, using $\Gamma(2-\alpha) = (1-\alpha)\Gamma(1-\alpha)$ we can check directly that $w_i^n = \mu([\theta_{i-1}^n, \theta_i^n])$: both sides simplify to $\frac{r_n^{1-\alpha}-1}{\Gamma(\alpha)\Gamma(2-\alpha)} r_n^{(1-\alpha)(i-1-n/2)}$ once the exponent in the definition of w_i^n is collected as $(\alpha-1)(1+n/2) + (1-\alpha)i = (1-\alpha)(i-1-n/2)$. Similarly, since $\nu(d\beta) = \beta^{1-\alpha} d\beta / (\Gamma(\alpha)\Gamma(1-\alpha))$ and $\Gamma(3-\alpha) = (2-\alpha)(1-\alpha)\Gamma(1-\alpha)$,

$$\nu([\theta_{i-1}^n, \theta_i^n]) = \frac{1-\alpha}{\Gamma(\alpha)\Gamma(3-\alpha)} \left((\theta_i^n)^{2-\alpha} - (\theta_{i-1}^n)^{2-\alpha} \right).$$

Substituting the explicit formulas for w_i^n and β_i^n (equation (2.6)), the product $w_i^n \beta_i^n$ has r_n -exponent $(1-\alpha)(i-1-n/2) + (i-1-n/2) = (2-\alpha)(i-1-n/2)$, and collecting constants shows $w_i^n \beta_i^n$ equals exactly the right-hand side above. $\square$

This is reassuring since β_i^n was originally defined (Section 2.1) as the μ -weighted centroid of its block ($w_i^n \beta_i^n = \int_{\theta_{i-1}^n}^{\theta_i^n} \beta \mu(d\beta)$) by construction - Lemma C.1.1 simply re-expresses this defining property in terms of ν . It follows that $-K''(t) := \sum_{i=1}^n w_i^n \beta_i^n e^{-\beta_i^n t} = \sum_{i=1}^n \nu([\theta_{i-1}^n, \theta_i^n]) e^{-\beta_i^n t}$ is *itself* an interval-mass quadrature approximation of $-K'$ with respect to ν , of exactly the same type as K^n is of K with respect to μ . This lets us reuse the three-piece argument of Theorem 2.2.1 essentially unchanged, one level up.

Lemma C.1.2 (Derivative kernel convergence). Let $K''(t) := \sum_{i=1}^n w_i^n (-\beta_i^n) e^{-\beta_i^n t}$. Under hypothesis (2.5) ($r_n \downarrow 1$, $n \log r_n \rightarrow \infty$),

$$\|K'' - K'\|_{L^2(1,\infty)} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

¹These are the same points denoted η_i^n in the proof of Theorem 2.2.1; we use θ_i^n here for consistency within this appendix.

Proof. As above, we write

$$K^{n'}(t) - K'(t) = \underbrace{\int_0^{\theta_0^n} e^{-\beta t} \nu(d\beta)}_{J_1(t)} + \underbrace{\sum_{i=1}^n \int_{\theta_{i-1}^n}^{\theta_i^n} (e^{-\beta_i^n t} - e^{-\beta t}) \nu(d\beta)}_{J_2(t)} + \underbrace{\int_{\theta_n^n}^{\infty} e^{-\beta t} \nu(d\beta)}_{J_3(t)},$$

the same low-frequency, mid-range and high-frequency split as in the proof of Theorem 2.2.1, now applied to ν in place of μ . We bound each in $L^2(1, \infty)$ restricting to $t \geq 1$, rather than $t \in (0, T)$, is what buys us the extra decay we need, since it keeps us away from the singularity of K' at $t = 0$.

Bounding J_1 . Substituting $\beta = x/t$,

$$|J_1(t)| \leq C \int_0^{\theta_0^n} \beta^{1-\alpha} e^{-\beta t} d\beta = C t^{\alpha-2} \int_0^{\theta_0^n t} x^{1-\alpha} e^{-x} dx \leq C t^{\alpha-2} \min(\Gamma(2-\alpha), (\theta_0^n t)^{2-\alpha}),$$

using the incomplete-Gamma-type bound $\int_0^a x^{1-\alpha} e^{-x} dx \leq \min(\Gamma(2-\alpha), a^{2-\alpha})$ (the second bound from simply dropping $e^{-x} \leq 1$ on a short interval). Splitting the integral $t \geq 1$ at the crossover point $t^* := 1/\theta_0^n$ - using the second bound for $t \in [1, t^*]$ and the first for $t \in [t^*, \infty)$ - both pieces are found to scale identically as $(\theta_0^n)^{3-2\alpha}$, giving

$$\int_1^\infty J_1(t)^2 dt \leq C' (\theta_0^n)^{3-2\alpha}.$$

For $\alpha \in (\frac{1}{2}, 1)$ we have $3-2\alpha \in (1, 2)$, so this exponent is strictly positive since $\theta_0^n = r_n^{-n/2} \rightarrow 0$ as $n \rightarrow \infty$ (a consequence of hypothesis (2.5)), $\|J_1\|_{L^2(1, \infty)} \rightarrow 0$.

Bounding J_3 . For $\beta \geq \theta_n^n =: \Theta_n$,

$$|J_3(t)| \leq e^{-\Theta_n t/2} \int_0^\infty \beta^{1-\alpha} e^{-\beta t/2} d\beta = C e^{-\Theta_n t/2} t^{\alpha-2},$$

so $\int_1^\infty J_3(t)^2 dt \leq C^2 e^{-\Theta_n} / \Theta_n \rightarrow 0$, since $\Theta_n = r_n^{n/2} \rightarrow \infty$ under the same hypothesis.

Bounding J_2 . By the mean value theorem, for $\beta \in [\theta_{i-1}^n, \theta_i^n]$,

$$|e^{-\beta_i^n t} - e^{-\beta t}| \leq t e^{-\theta_{i-1}^n t} (\theta_i^n - \theta_{i-1}^n),$$

and $\nu([\theta_{i-1}^n, \theta_i^n]) \leq C (\theta_{i-1}^n)^{2-\alpha} (r_n - 1)$ for n sufficiently large (using $\theta_i^n - \theta_{i-1}^n = \theta_{i-1}^n (r_n - 1)$ together with the mean value theorem applied to $x \mapsto x^{2-\alpha}$). Hence

$$|J_2(t)| \leq C t (r_n - 1)^2 \sum_{i=1}^n (\theta_{i-1}^n)^{3-\alpha} e^{-\theta_{i-1}^n t} =: C t (r_n - 1)^2 S_n(t).$$

Writing $\theta_{i-1}^n = (\theta_i^n - \theta_{i-1}^n)/(r_n - 1)$ turns $S_n(t)$ into a left Riemann sum $\frac{1}{r_n-1} \sum_i h(\theta_{i-1}^n) (\theta_i^n - \theta_{i-1}^n)$ for $h(\beta) := \beta^{2-\alpha} e^{-\beta t}$. Since h is bell shaped,

this sum approximates $\int_0^\infty h d\beta$ up to a factor bounded by $(1 + r_n)$ (a standard comparison for a left Riemann sum against a nearly uniform geometric partition), giving $\sum_i h(\theta_{i-1}^n)(\theta_i^n - \theta_{i-1}^n) \leq (1 + r_n) \Gamma(3 - \alpha) t^{\alpha-3}$. Thus $S_n(t) \leq C t^{\alpha-3}/(r_n - 1)$, and therefore

$$|J_2(t)| \leq C(r_n - 1) t^{\alpha-2}, \quad \|J_2\|_{L^2(1,\infty)}^2 \leq C^2(r_n - 1)^2 \int_1^\infty t^{2\alpha-4} dt \rightarrow 0,$$

using only $r_n \downarrow 1$ and, for $\alpha \in (\frac{1}{2}, 1)$, $2\alpha - 4 \in (-3, -2)$, so in particular $2\alpha - 4 < -1$ and the improper integral converges.

Combining the three bounds, $\|K^{n'} - K'\|_{L^2(1,\infty)} \leq \|J_1\|_{L^2(1,\infty)} + \|J_2\|_{L^2(1,\infty)} + \|J_3\|_{L^2(1,\infty)} \rightarrow 0$. $\square$

C.2 From derivative convergence to the compensator bound

With Lemma C.1.2 in hand, we can now return to $R_n(t)$ and close the argument. The idea is to convert the difference $\Delta^n(t + w) - \Delta^n(w)$ into an integral of $(\Delta^n)' = K^{n'} - K'$ via the fundamental theorem of calculus, then bound the resulting expression using Cauchy-Schwarz and Tonelli's theorem - trading a difference of a badly-behaved kernel for an integral of a well-behaved one.

Theorem C.2.1 ($L^2(\Omega)$ convergence of lifted fBm to Mandelbrot–Van Ness fBm). *Under hypothesis (2.5), for every $T > 0$,*

$$\sup_{t \in [0, T]} \mathbb{E}[(B_t^{H,n} - B_t^H)^2] \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Proof. As recalled at the start of this appendix, $\mathbb{E}[(B_t^{H,n} - B_t^H)^2] = A_n(t) + R_n(t)$, and $A_n(t) \leq \|K^n - K\|_{L^2(0,T)}^2$ for $t \in [0, T]$, so $\sup_{t \in [0, T]} A_n(t) \rightarrow 0$ directly by Theorem 2.2.1.

For $R_n(t)$, split the integral in (C.1) at $w = 1$:

$$R_n(t) = \underbrace{\int_0^1 (\Delta^n(t + w) - \Delta^n(w))^2 dw}_{R_n^{\text{near}}(t)} + \underbrace{\int_1^\infty (\Delta^n(t + w) - \Delta^n(w))^2 dw}_{R_n^{\text{far}}(t)}.$$

Near piece. Bounding as before via $(a - b)^2 \leq 2a^2 + 2b^2$, both $t + w$ and w range over $[0, T + 1]$ for $t \in [0, T]$, $w \in [0, 1]$, so $R_n^{\text{near}}(t) \leq 4\|K^n - K\|_{L^2(0, T+1)}^2 \rightarrow 0$ by Theorem 2.2.1 once more.

Far piece. Here we start with the fundamental theorem of calculus,

$$\Delta^n(t + w) - \Delta^n(w) = \int_0^t (\Delta^n)'(w + s) ds.$$

By Cauchy-Schwarz on the s -integral over $[0, t]$,

$$(\Delta^n(t + w) - \Delta^n(w))^2 \leq t \int_0^t ((\Delta^n)'(w + s))^2 ds.$$

Integrating over $w \in [1, \infty)$ and applying Tonelli's theorem to swap the order of integration,

$$R_n^{\text{far}}(t) \leq t \int_0^t \int_1^\infty ((\Delta^n)'(w+s))^2 dw ds.$$

For each fixed $s \in [0, t]$, substituting $w' = w + s \geq 1 + s \geq 1$ shows the inner integral is bounded by $\|(\Delta^n)'\|_{L^2(1, \infty)}^2$ (the domain $[1 + s, \infty)$ is contained in $[1, \infty)$), independent of s . Hence

$$R_n^{\text{far}}(t) \leq t \int_0^t \|(\Delta^n)'\|_{L^2(1, \infty)}^2 ds = t^2 \|K^{n'} - K'\|_{L^2(1, \infty)}^2,$$

and taking the supremum over $t \in [0, T]$,

$$\sup_{t \in [0, T]} R_n^{\text{far}}(t) \leq T^2 \|K^{n'} - K'\|_{L^2(1, \infty)}^2,$$

which vanishes as $n \rightarrow \infty$ by Lemma C.1.2.

Combining all four bounds $A_n(t)$, $R_n^{\text{near}}(t)$, and $R_n^{\text{far}}(t)$, each vanishing uniformly over $t \in [0, T]$ we get that

$$\sup_{t \in [0, T]} \mathbb{E}[(B_t^{H, n} - B_t^H)^2] \rightarrow 0 \quad \text{as } n \rightarrow \infty. \quad \square$$

Theorem C.2.1 closes the gap left open in Section 2.2: convergence of lifted fBm to true, stationary-increment (Mandelbrot–Van Ness) fBm now holds unconditionally under hypothesis (2.5). The essential idea - that the compensator term, though not directly amenable to the kernel-convergence bound, becomes tractable after differentiating and working away from the origin.

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